



Gene Therapy

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Objectives

Explain the different approaches to gene therapy

Enumerate diseases that can be treated with gene therapy

Compare and contrast between the different gene delivery systems

Outline

Definition of gene therapy

Goal of gene therapy

Gene therapy techniques

Requirements for gene therapy

Somatic vs germline gene therapy

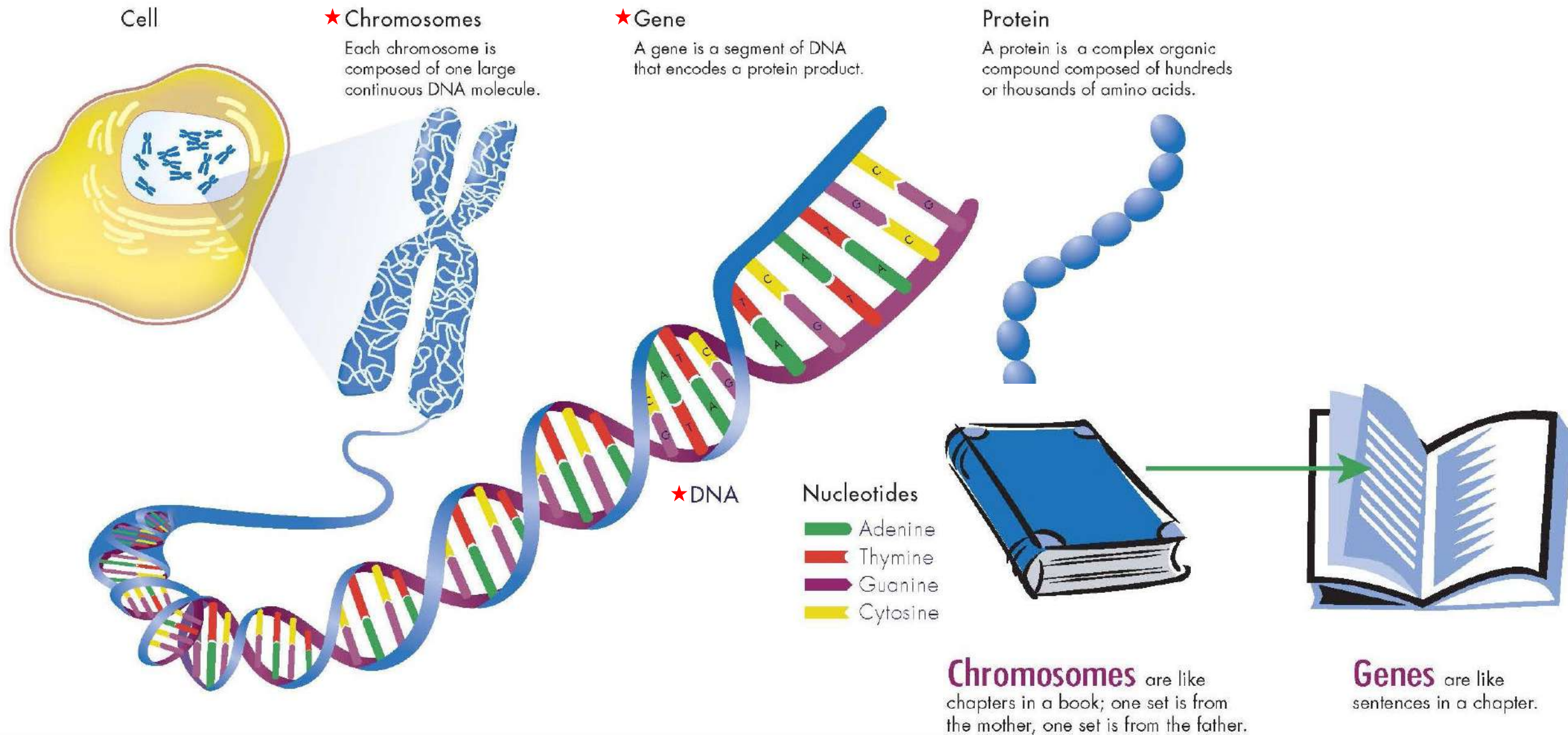
Strategies for gene transfer

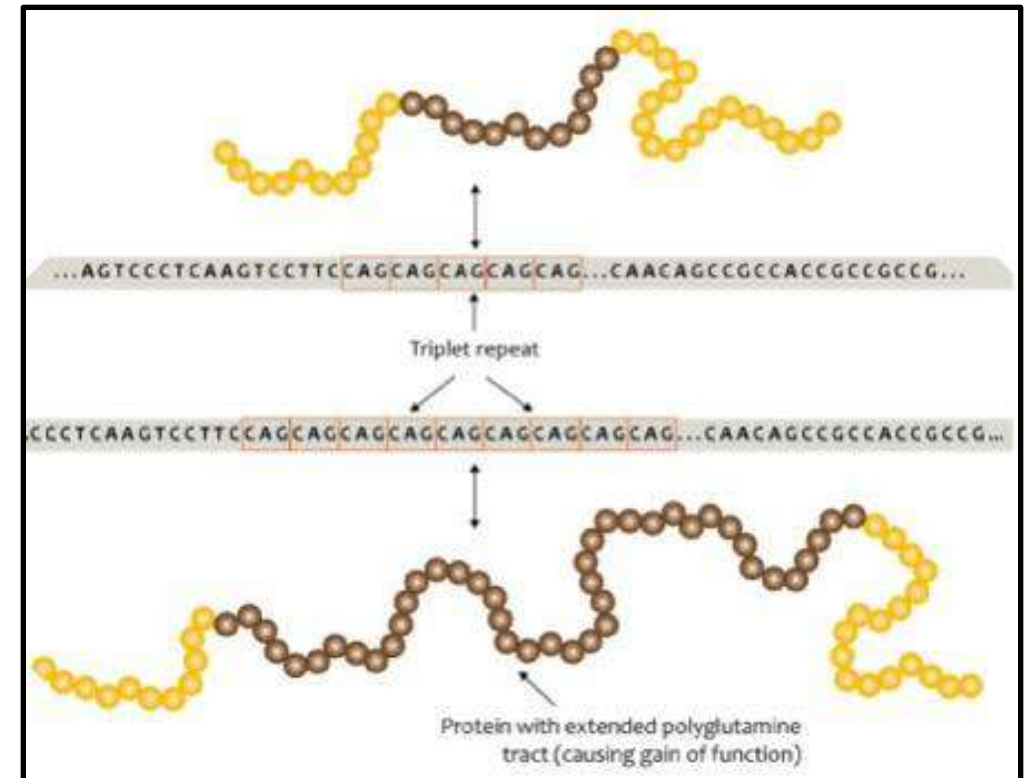
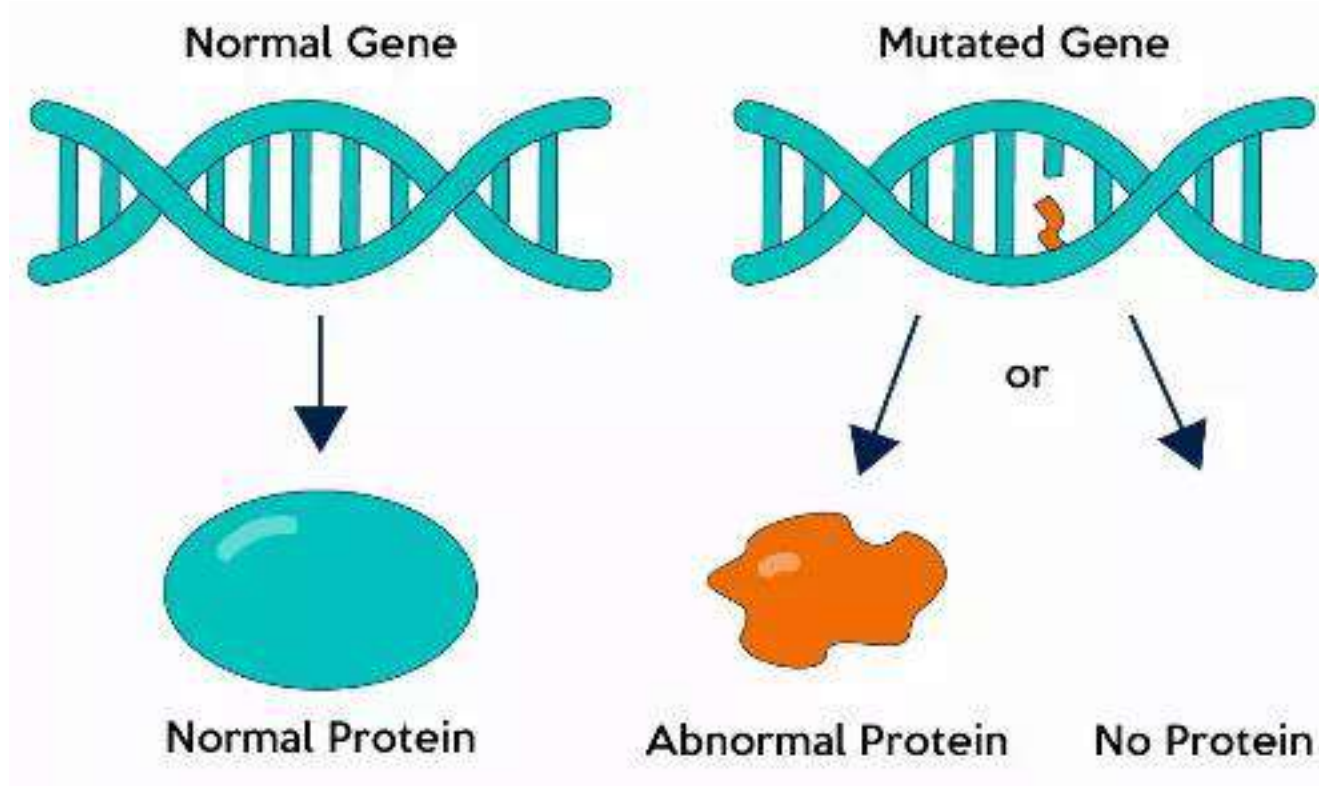
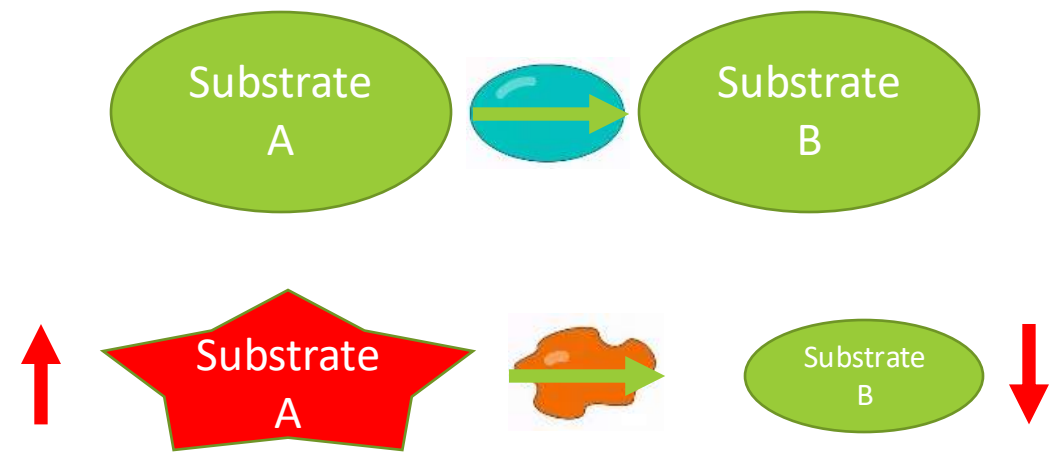
Vectors used in gene therapy

Examples of disorders can be treated by gene therapy



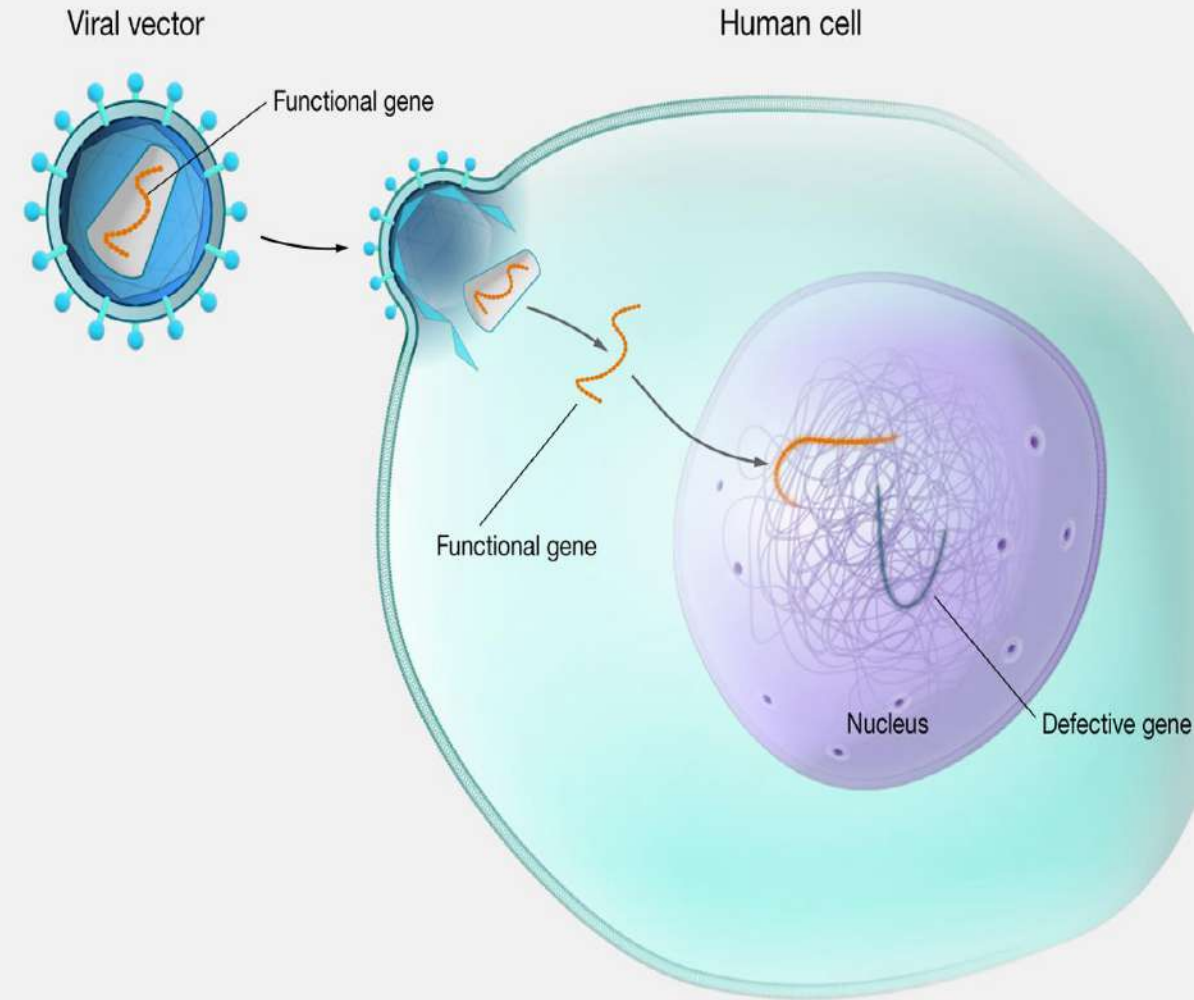
Chromosome to Gene to Protein

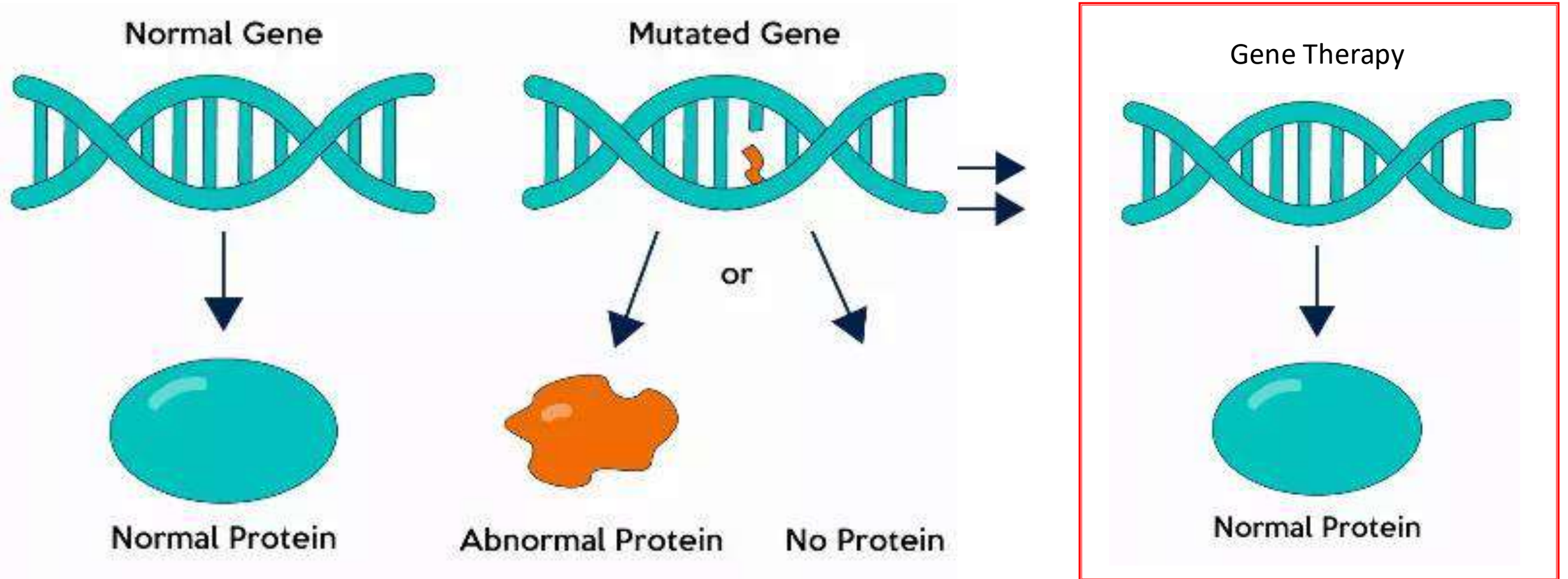




Definition of Gene Therapy

- Gene therapy is a technique that uses genetic material in the treatment or prevention of disease*
- Both inherited (e.g., hemophilia and sickle cell disease) and acquired (e.g., leukemia) disorders have been treated with gene therapy



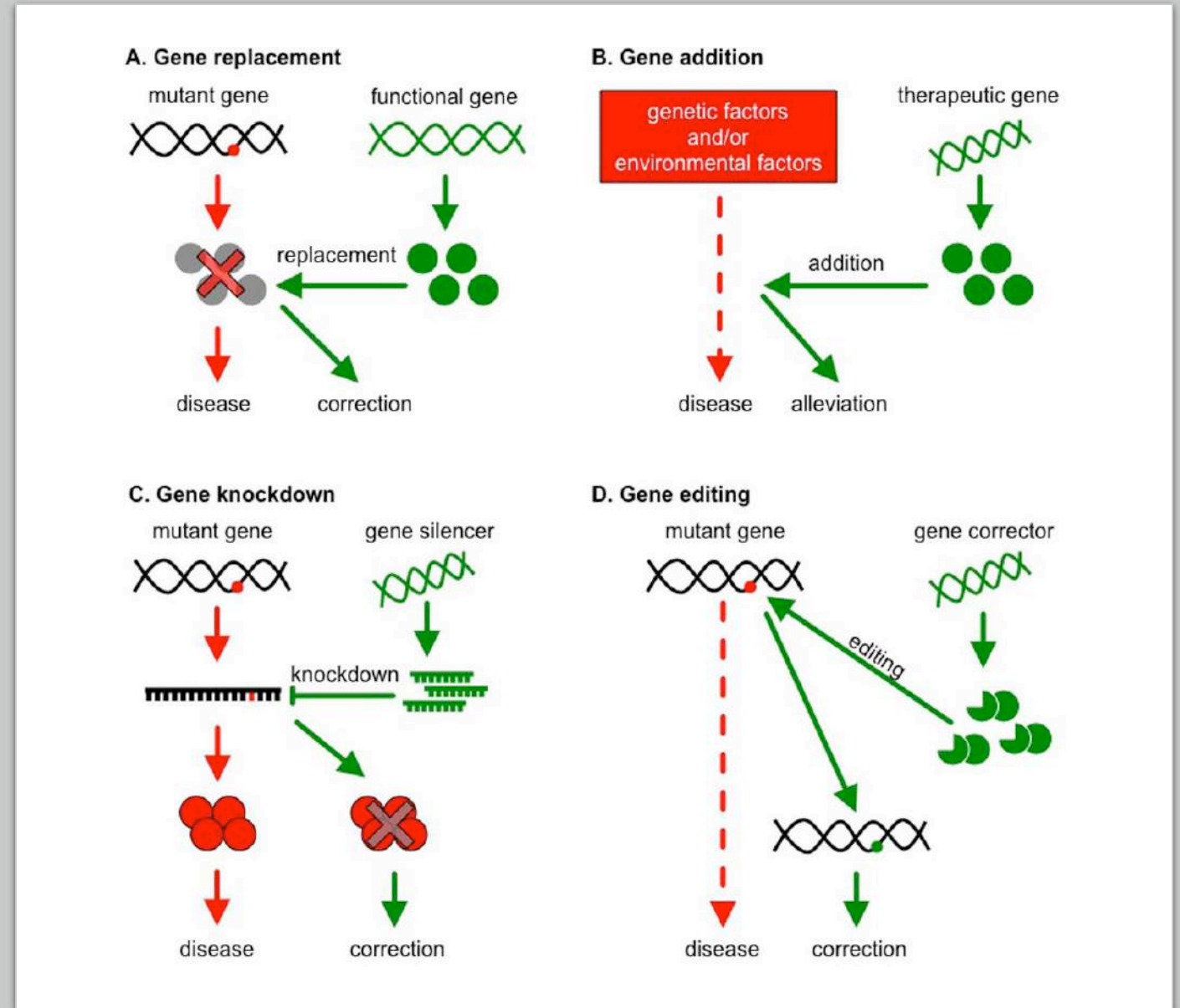


Goal of Gene Therapy?

- To prevent the pathogenetic events that damage cells and correct the reversible features
- To repair or change a gene, or treat a disease, based on pathogenesis:
 - Reduce levels of a disease-causing version of a protein
 - Increase production of disease-fighting proteins
 - Produce new/modified proteins

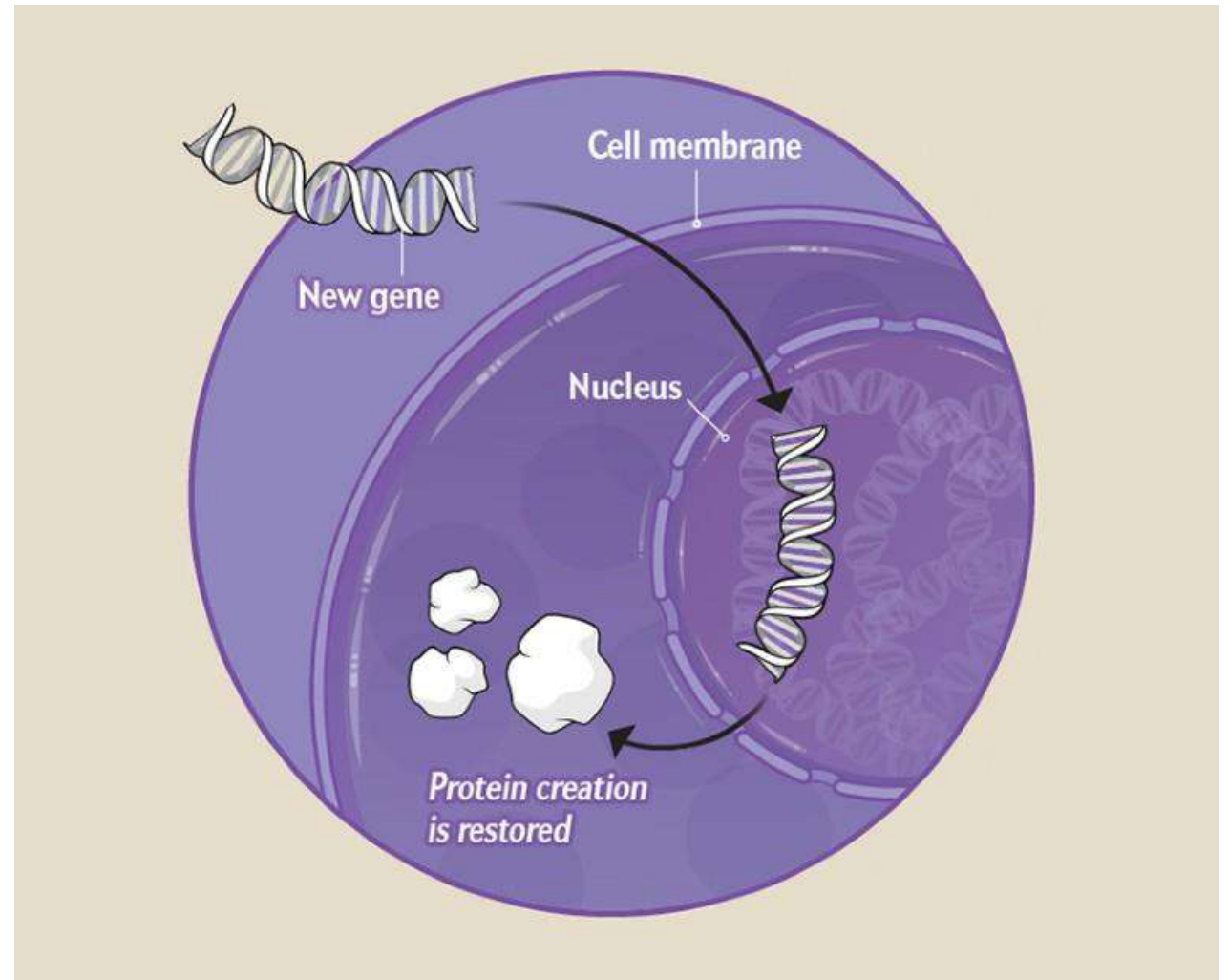
Gene Therapy Techniques

- Gene replacement (introducing a new gene)
- Gene augmentation (modifying an existing gene's activity)
- Gene silencing (gene knockdown)
- Gene editing (edit gene directly)



Gene Replacement

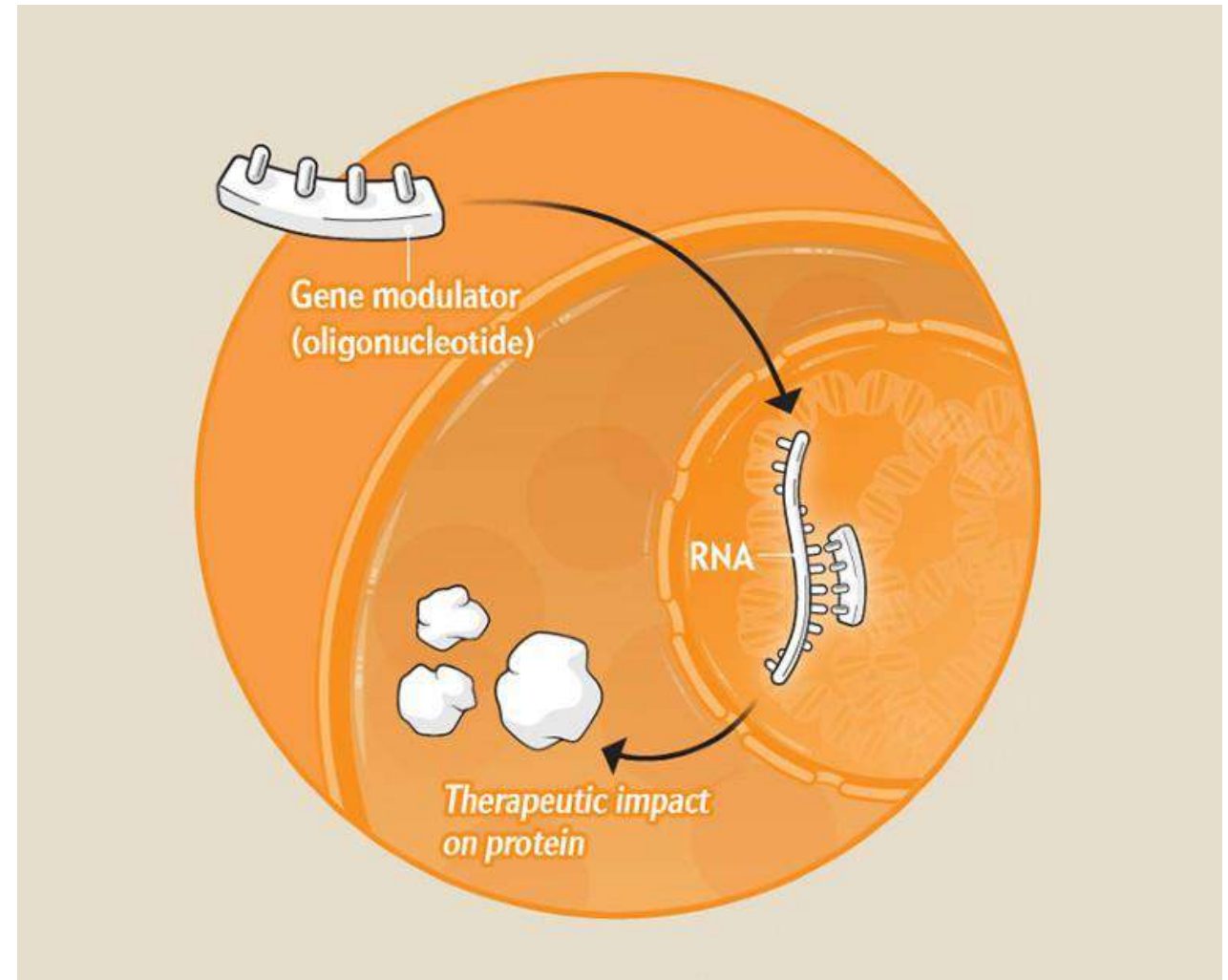
- Introducing a new gene
- Includes removal of a mutant gene and replacement with a normal, functional gene (loss of function genetic disease e.g., DMD)



Landhuis E. The Definition of Gene Therapy Has Changed. Nature. 2021 Oct 26. doi: 10.1038/d41586-021-02736-8. Epub ahead of print. PMID: 34703017.

Gene Augmentation

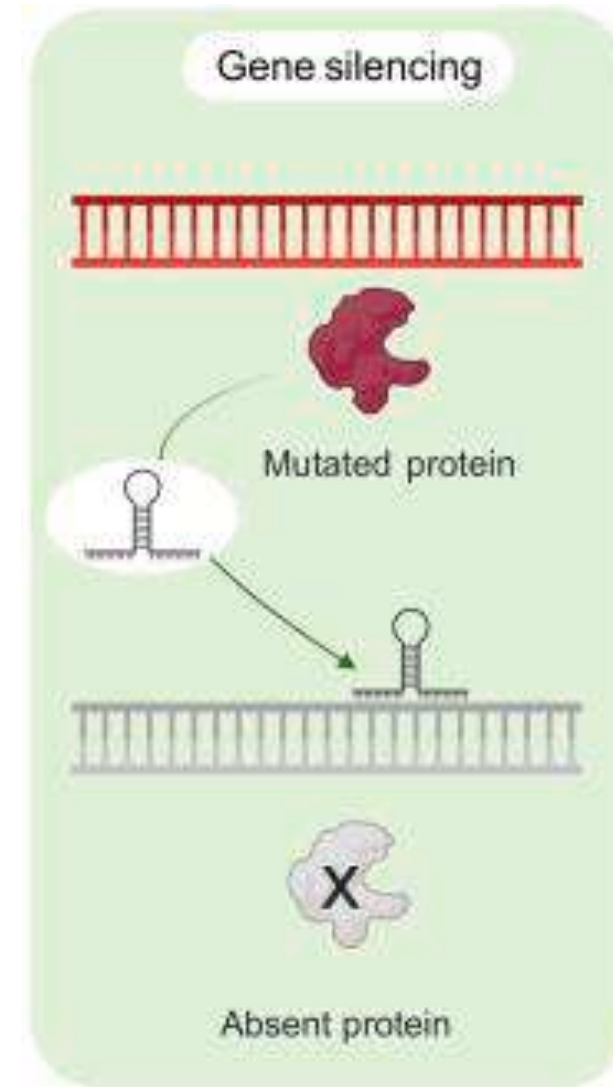
- Modifying an existing gene's activity
- Modifying the content or expression by introducing short sequences of nucleic acids, oligonucleotides, to impact disease state
- Alleviating the disease by supplementing an alternative gene that target a specific aspect of the disease mechanism



Landhuis E. The Definition of Gene Therapy Has Changed. Nature. 2021 Oct 26. doi: 10.1038/d41586-021-02736-8. Epub ahead of print. PMID: 34703017.

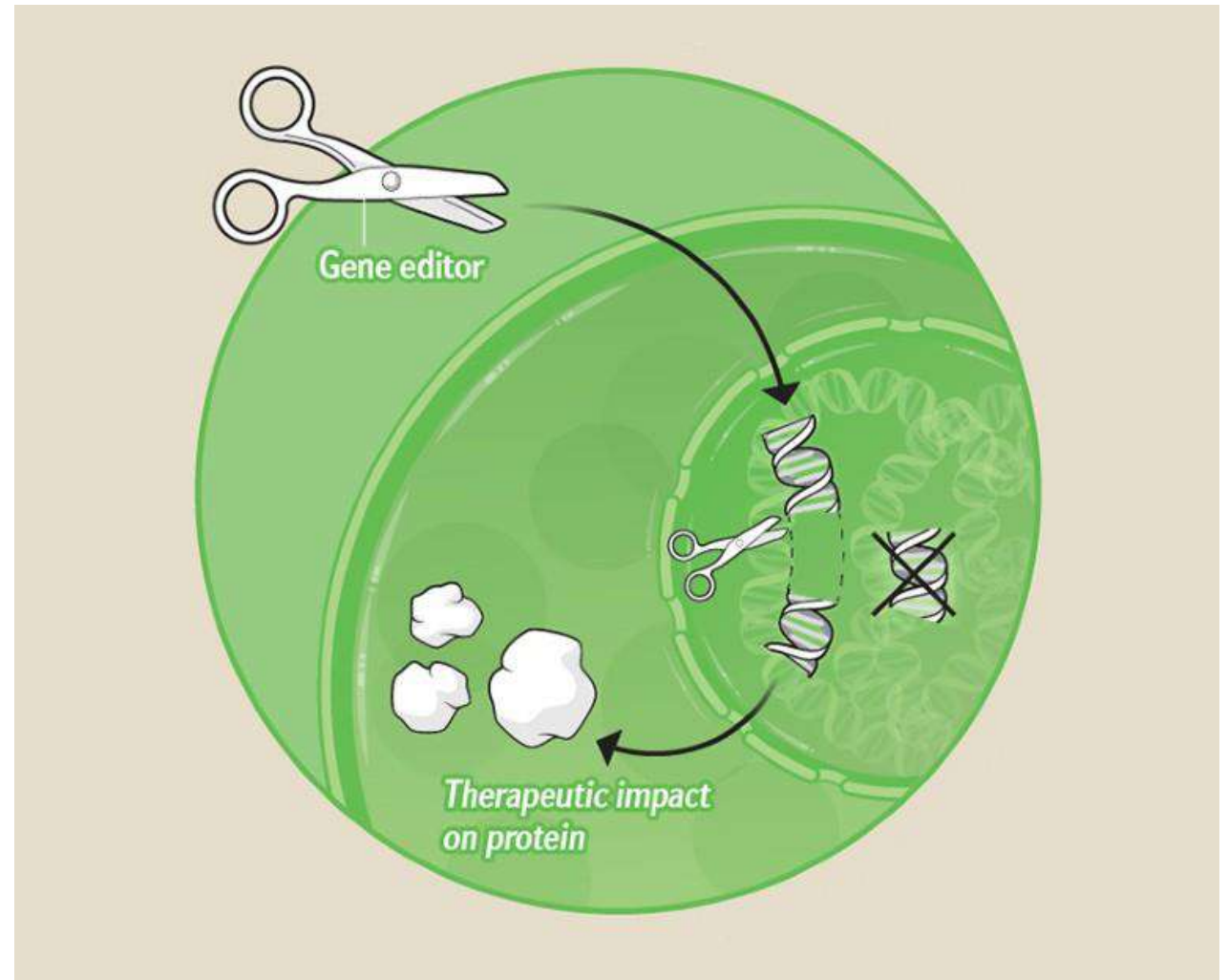
Gene Silencing

- Gene knockdown
- Inactivation of a mutated gene that is toxic to cells (gain of function genetic disorders; e.g., Huntington disease)



Gene Editing

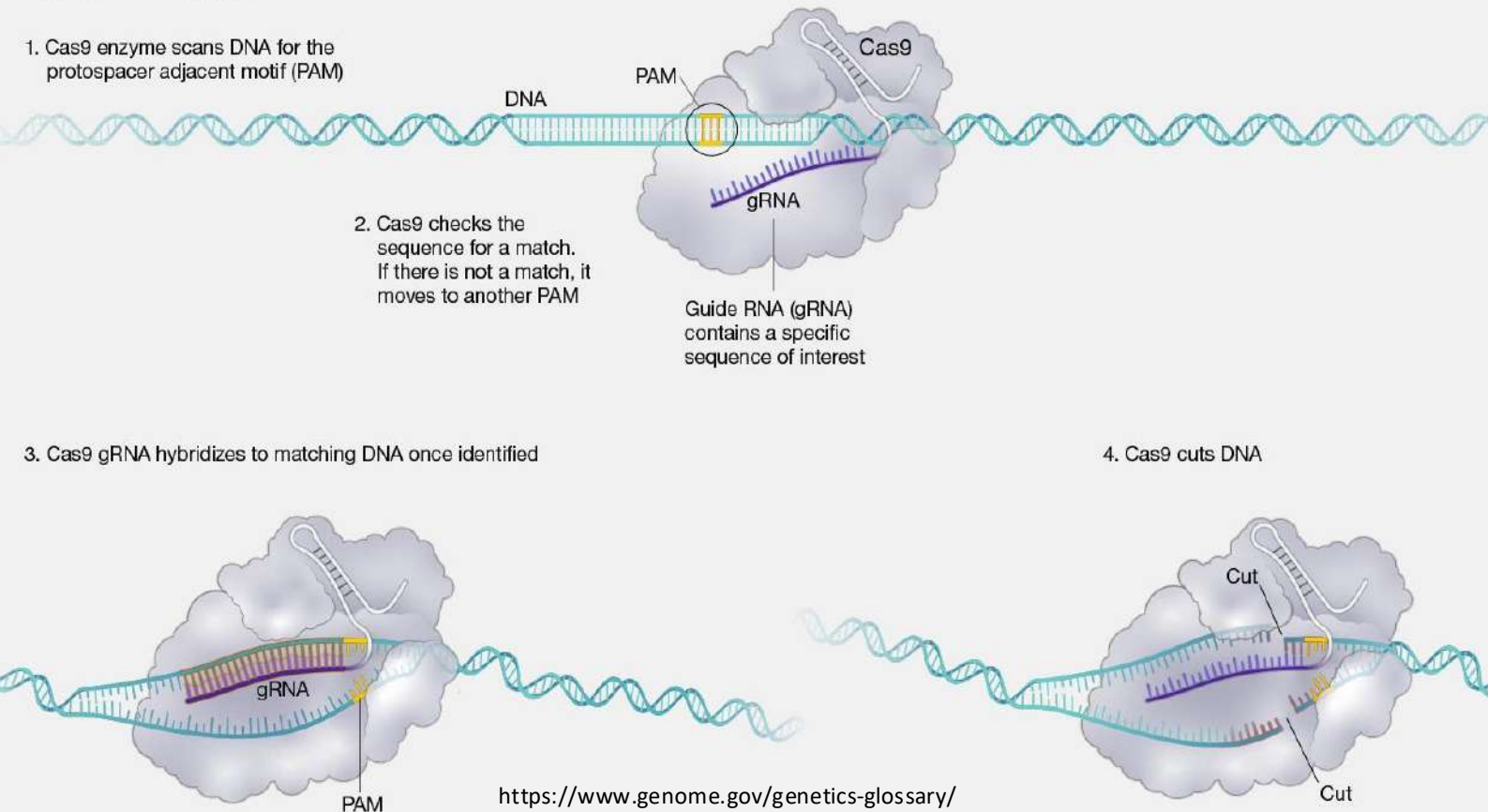
- Editing a gene directly
- By using gene-editing system, e.g., CRISPR-Cas9, to make precise changes in the genome



Landhuis E. The Definition of Gene Therapy Has Changed. *Nature*. 2021 Oct 26. doi: 10.1038/d41586-021-02736-8. Epub ahead of print. PMID: 34703017.

Gene Editing Using CRISPR-Cas9 (*Clustered, Regularly Interspaced, Short Palindromic Repeats and CRISPR-associated protein 9*)
AKA “Genetic Scissor”

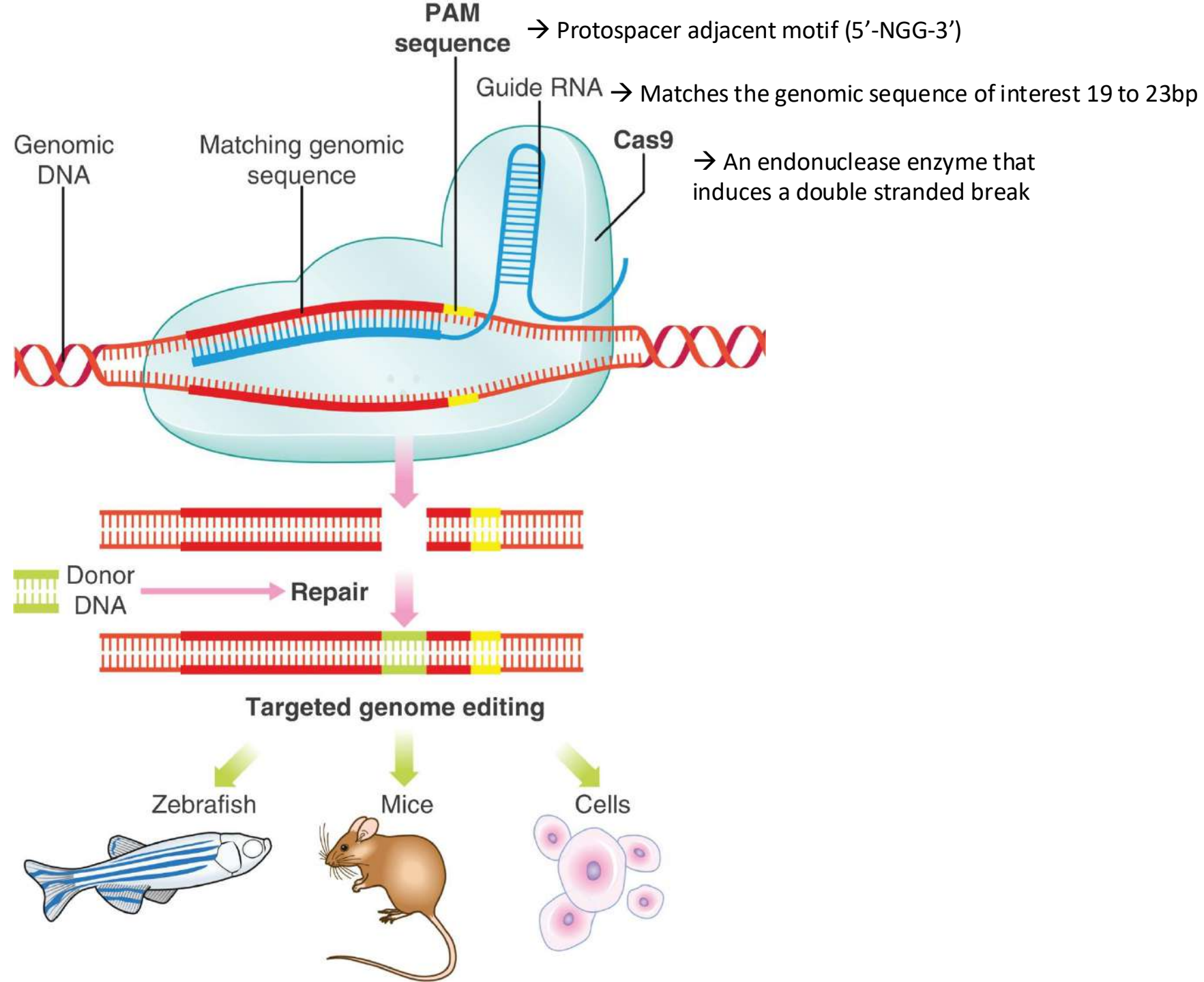
CRISPR-Cas9



- Discovered as a bacterial immune defense system that detects specific DNA seq of invading viruses and directs an enzyme to cut/destroy viral genome
- CRISPR-Cas9 is 2020 Nobel Prize-winning gene editing technology, it can:
 - Target a specific section of DNA
 - Makes a precise cut at the target site
 - Either to make a gene non-functional or replace it with another
- Can be used to edit nearly any location in the genome → great potential for treating disease

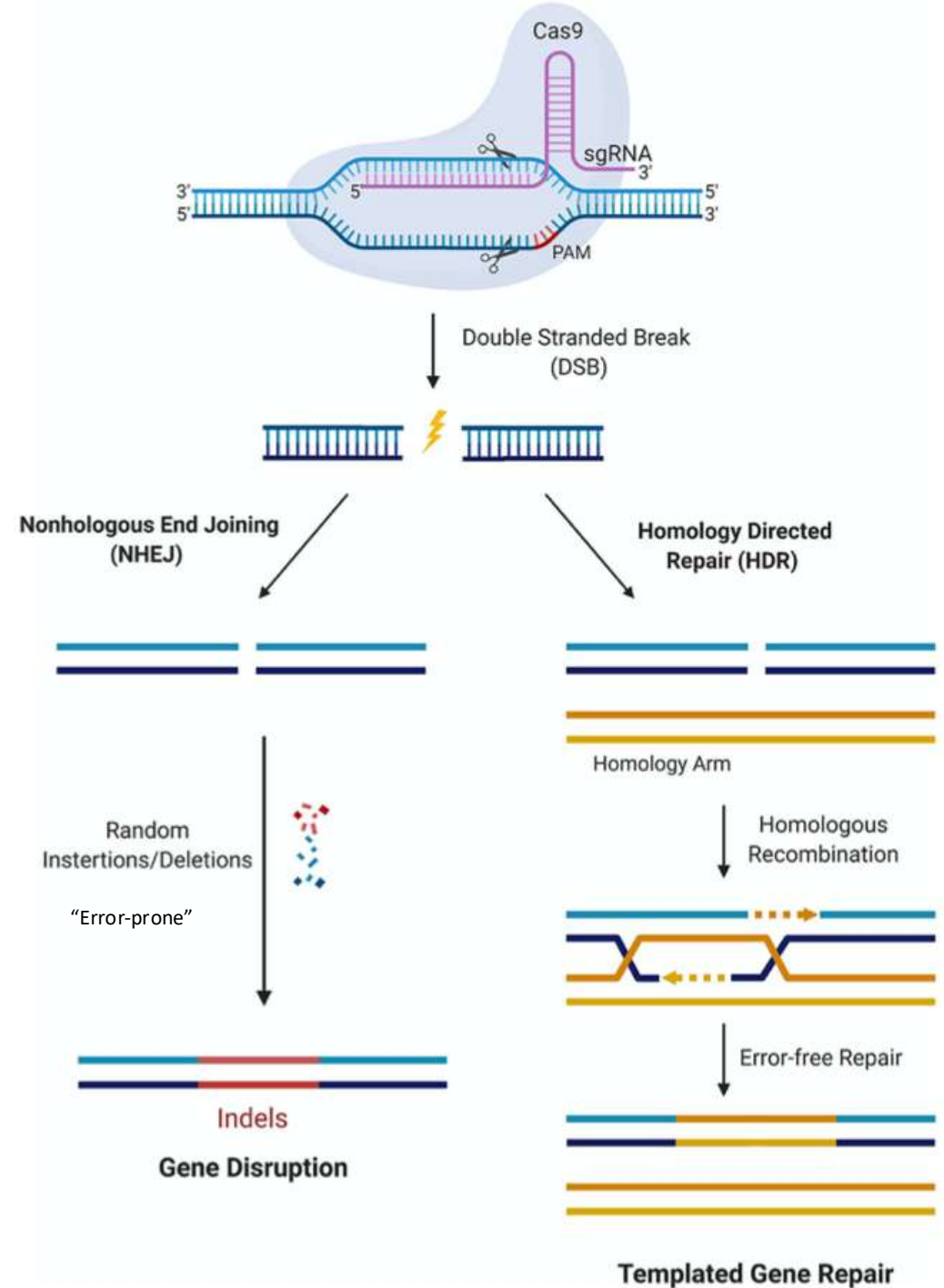
CRISPR/Cas9

Molecular scissors



How Does CRISPR-Cas9 Work?

Relies on either of two endogenous DNA repair pathways



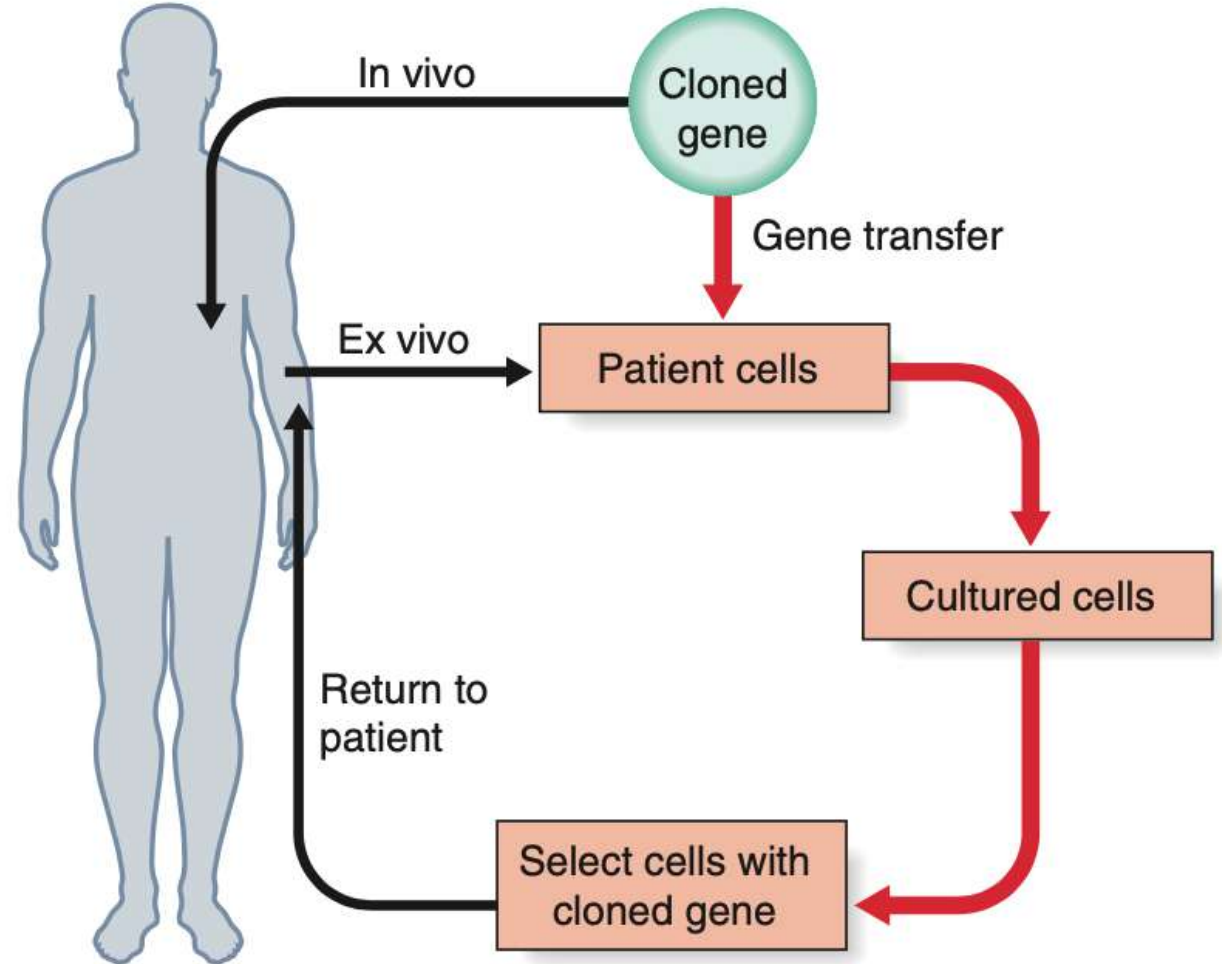
Strategies for Gene Transfer:



Ex-vivo gene therapy
transfer of genes to
cultured cells and
reinsertion



In-vivo gene therapy
direct delivery of
genes into the cells
of a particular tissue
in the body



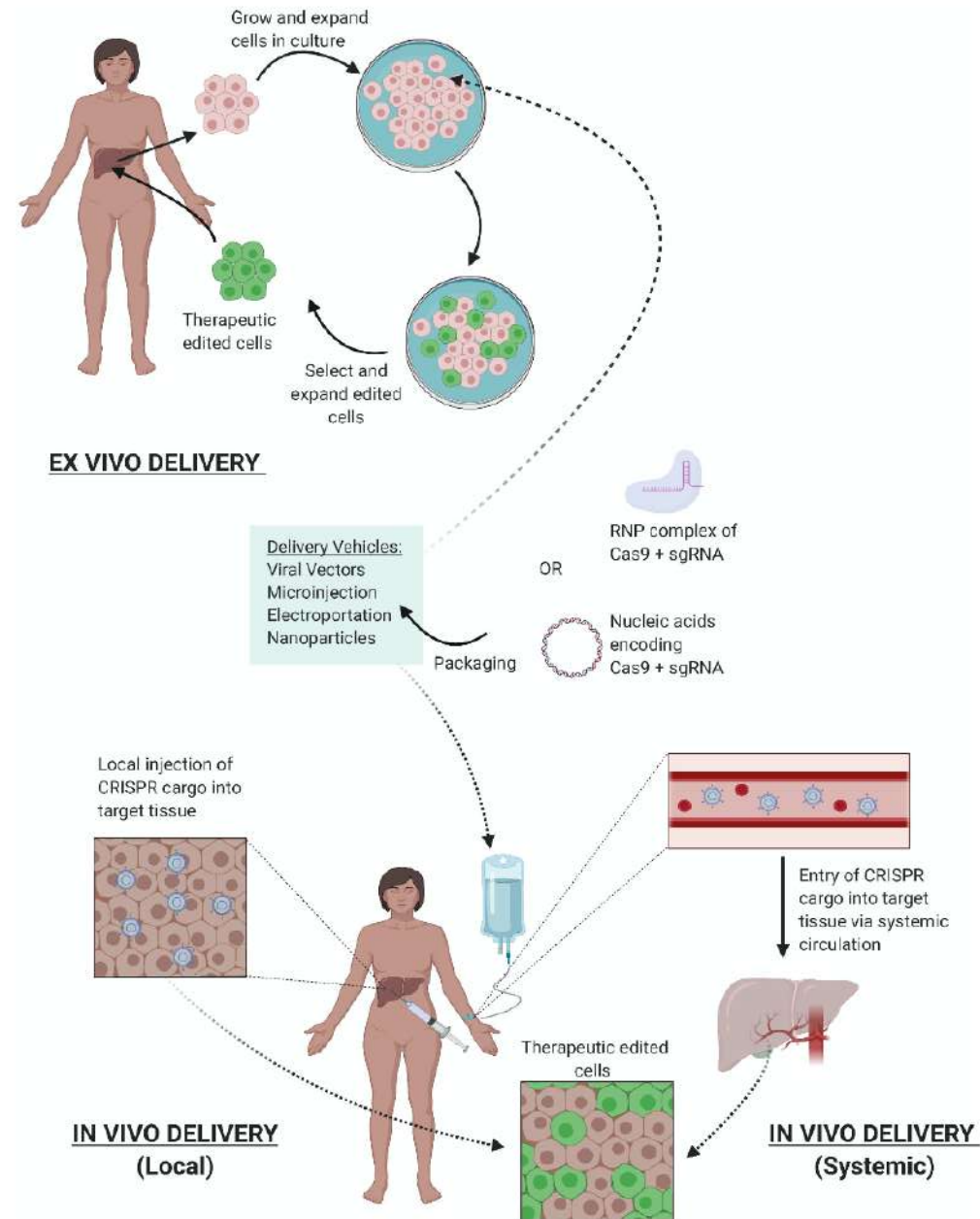
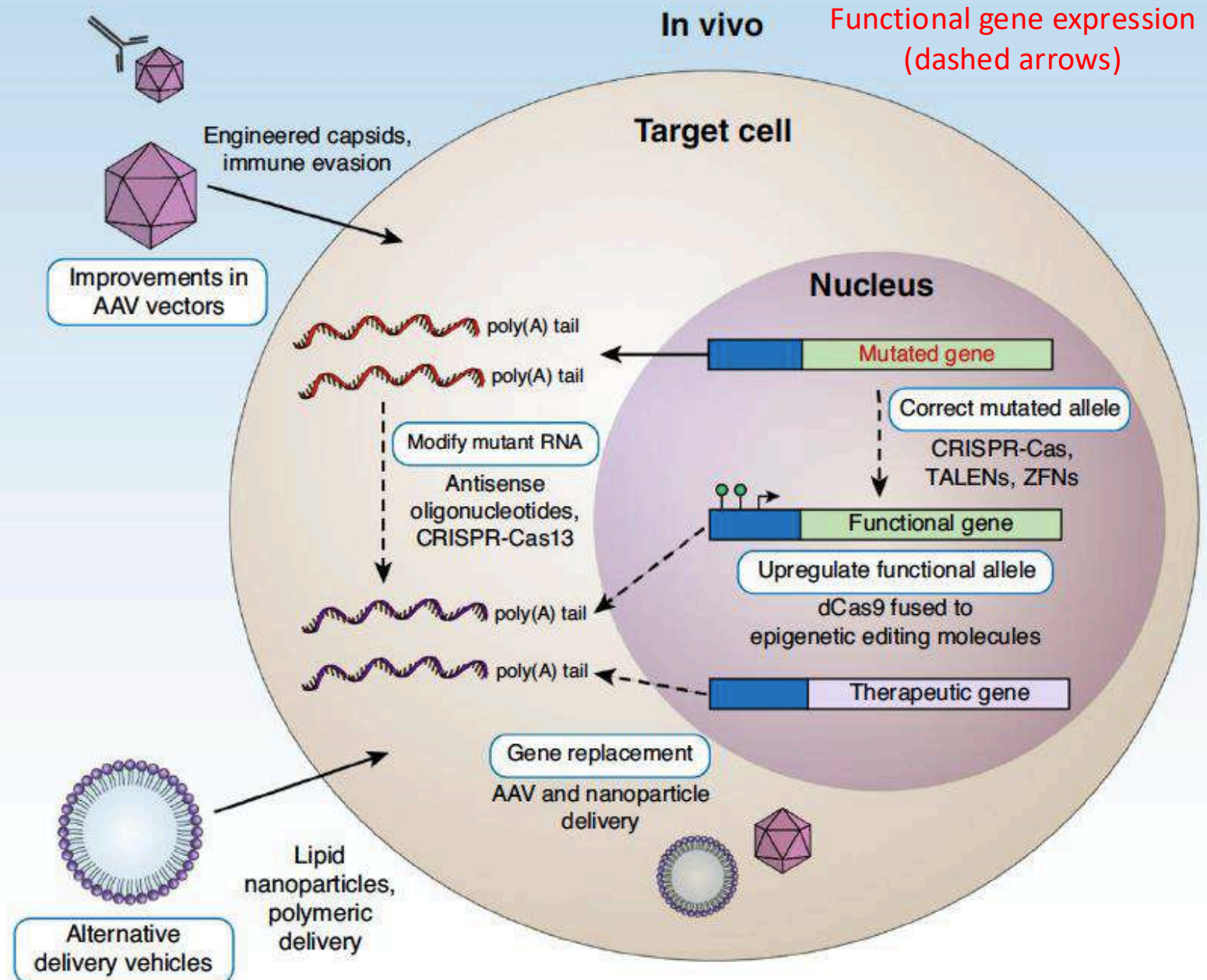
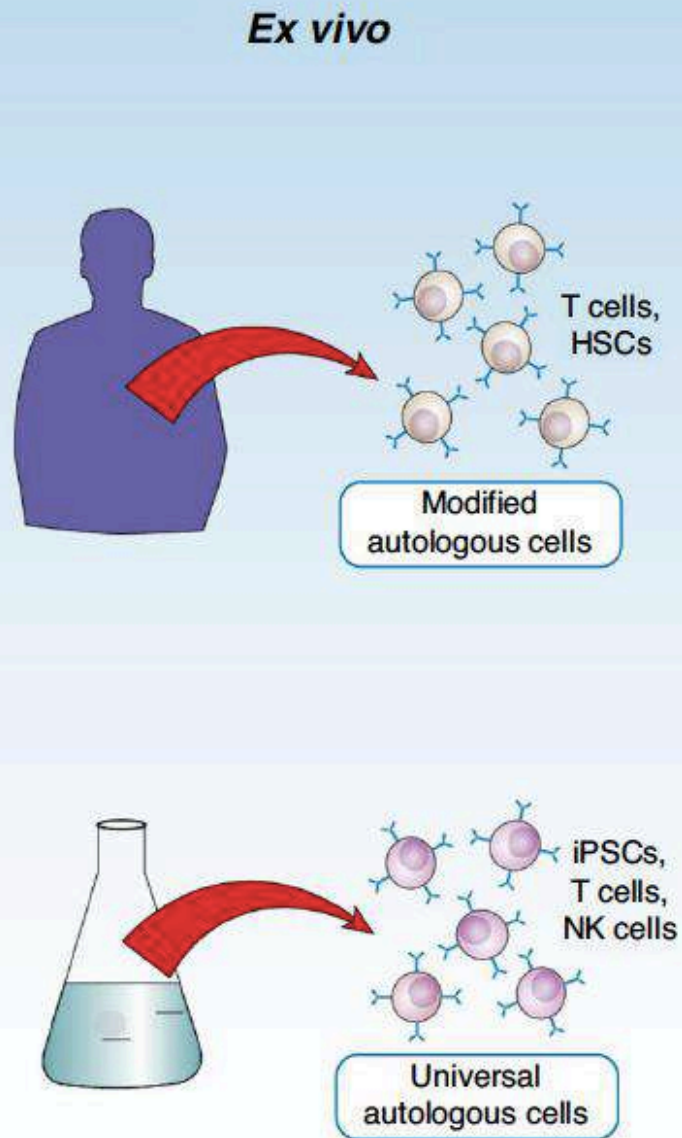


FIGURE 4 | Delivery of CRISPR Therapy. Nucleic acids encoding CRISPR/Cas9 or its RNP complex can be packaged into delivery vehicles. Once packaged, edits can be facilitated either *ex vivo* or *in vivo*. *Ex vivo* editing involves extraction of target cells from the patient, cell culture, and expansion *in vitro*, delivery of the CRISPR components to yield the desired edits, selection, and expansion of edited cells, and finally reintroduction of therapeutic edited cells into the patient. *In vivo* editing can be systemically delivered via intravenous infusions to the patient, where the CRISPR cargo travels through the bloodstream via arteries leading to the target tissue, or locally delivered with injections directly to target tissue. Once delivered, the edits are facilitated *in vivo* to provide therapeutic benefit.

Strategies for gene transfer:



Requirements for Gene Therapy

Disease

- Knowledge of pathophysiology
- Knowledge of molecular defect/gene
- Lack of good alternative therapy (Risk/benefit)

Regulation

- Safe and efficacious

Delivery

- An appropriate target cell.
 - Accessible
 - Long half-life/replicative potential
- Appropriate vector

Therapy

- Availability of functional copy of gene
- Availability of regulatory components

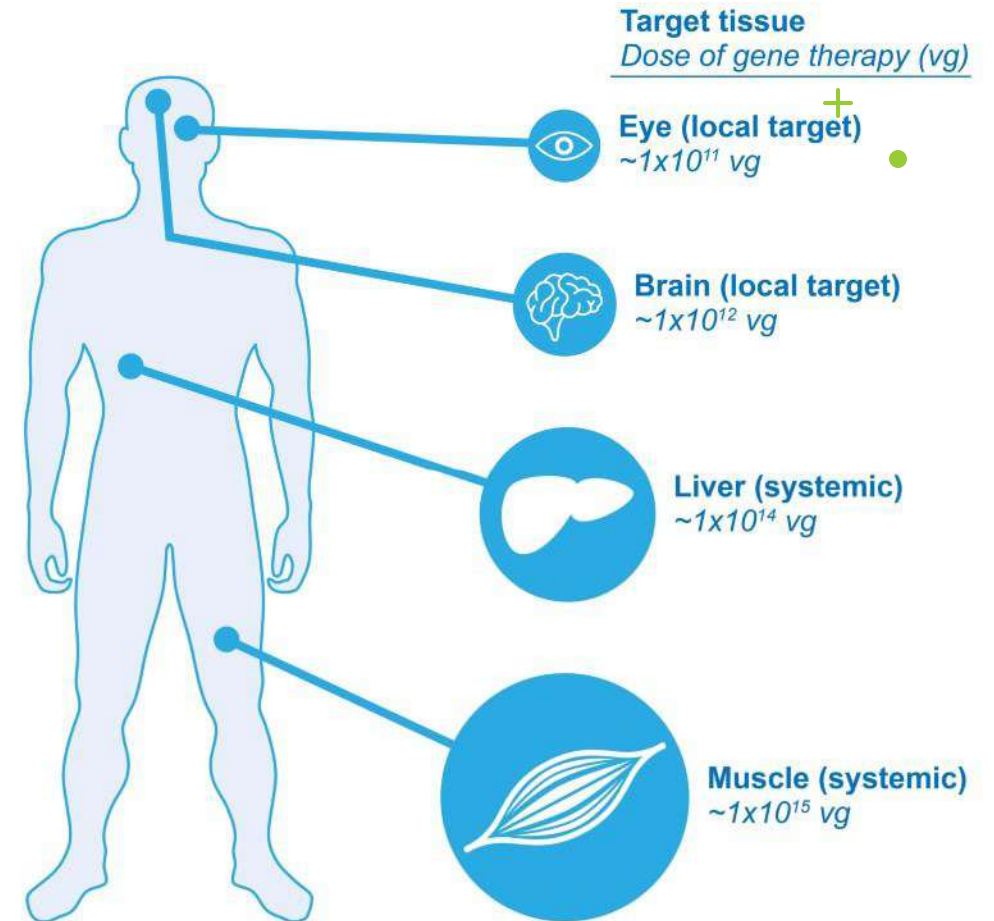
Target Cell

Which cells are ideal for transfer?

- Stem cell or progenitor cells
- Affected cells

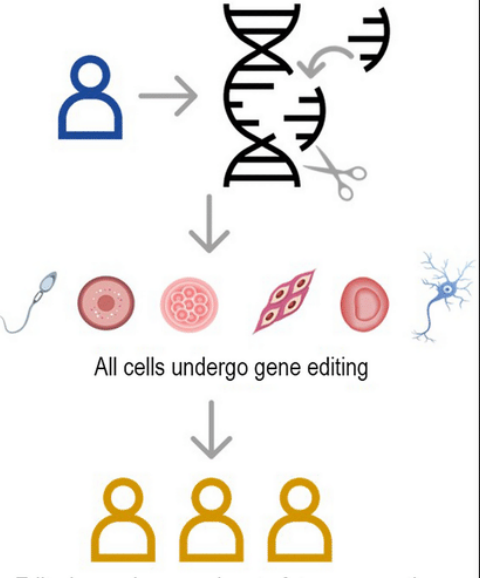
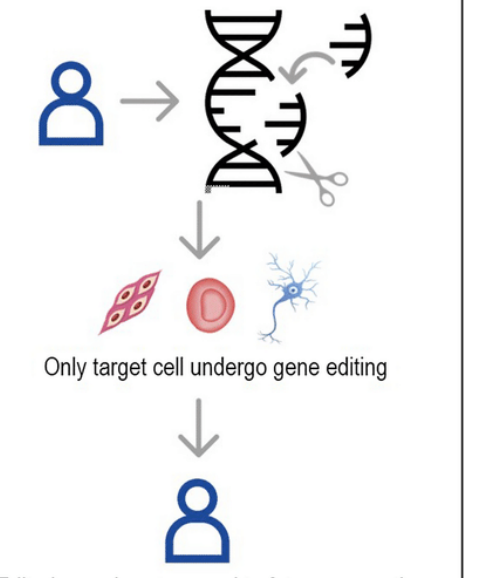
Number of cells required

- In PKU
 - 5% of the hepatocyte mass $\rightarrow \sim 10^{10}$ cells
 - Could be less if the expression of the gene is higher than the wildtype
- Muscular dystrophies
 - Needs to be inserted into a significant fraction of myocytes in order to have therapeutic efficacy



Tretiakova, A. P., Murphy, J. E., Binks, M., Mensah, P., Rabinowitz, J., McCarty, D. M., ... & Cheng, S. H. (2021). Realizing the promise of gene therapy through collaboration and partnering: Pfizer's view. *Nature Portfolio Sponsor Feature*
<http://www.nature.com/articles/d42473-018-00307-6>

Somatic vs Germline Gene Therapy

Germline gene editing	Somatic gene editing
Examples of cells	
 Sperm cell Ovum cell Early embryo	 Muscle cell Blood cell Nerve cell
Modification and effects	
 All cells undergo gene editing Edited gene is passed on to future generations	 Only target cell undergo gene editing Edited gene is not passed to future generations
Issues	
• Straightforward delivery of edited gene into reproductive cell • All cells will have functioning gene • Ethically controversial • Inconsistent and variable guidelines and regulation • Limited research/application permitted • Manipulation passed to next generation, resulting in good or bad outcomes	• Tricky delivery of edited gene into target cells • Difficult to achieve gene functioning in the genome • Therapy is short-lived and has to be repeated • Regulations available to govern applications • Research/application permitted as only actual person affected

Vectors

- **Ideally**
 - Safe
 - Readily made
 - Easily introduced into the appropriate target tissue
 - Would allow the expression of the gene of interest for life
- **Risks**
 - Immunotoxicity: harmful immune response to either the vector or transgene
 - Insertional mutagenesis causing malignancy
 - Insertional inactivation of an essential gene

Types of Vectors

Non-viral vectors



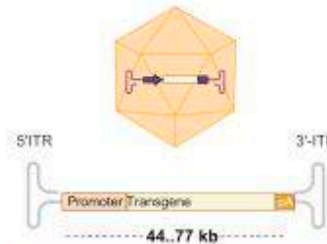
Cationic polymers



Lipid nanoparticles

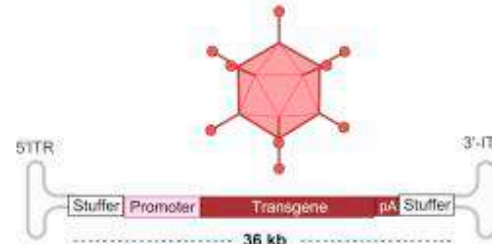
Easily scalable production
Lower costs of mass production compared to viral vectors
Packaging capacity: **unlimited size**
Low toxicity
Low immunogenicity
Poor efficiency to penetrating into the nucleus
Transient transgene expression

Viral vectors



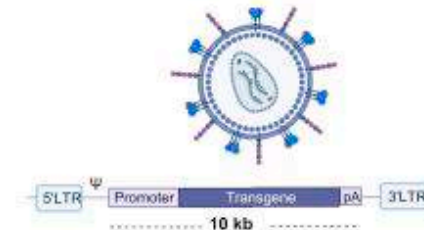
Adeno-associated virus (AAV)

ssDNA vector
Non-enveloped
Packaging capacity: **up to 4.7 kb**
Mainly **episomal**
Serotype-dependent pre-existing immunity
Low inflammatory responses
High liver transduction



HD-Adenovirus

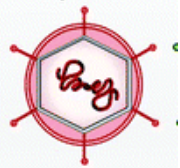
dsDNA vector
Non-enveloped
Packaging capacity: **up to 36 kb**
Mainly **episomal**
Serotype-dependent pre-existing immunity
Acute **inflammatory** responses
High liver transduction



Lentivirus

ssRNA vector
Enveloped
Packaging capacity: **up to 10 kb**
Integrative
Low pre-existing immunity
Poorly characterized in vivo
Ideal for HSCs transduction

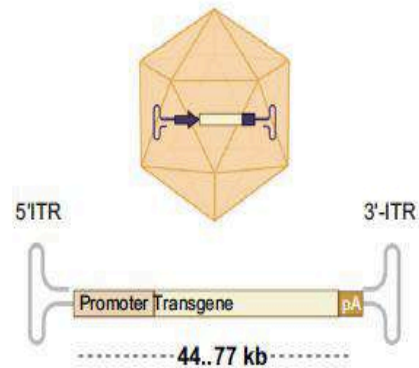
Herpes Virus



Herpes virus

dsDNA vector
Enveloped
Packaging capacity: 30-50 kb
Episomal
Neurotropic virus

Viral vectors



Adeno-associated virus (AAV)

ssDNA vector

Non-enveloped

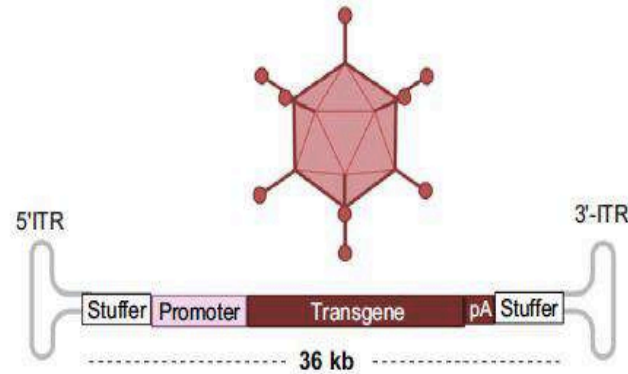
Packaging capacity: up to 4.7 kb

Mainly episomal

Serotype-dependent pre-existing immunity

Low inflammatory responses

High liver transduction



HD-Adenovirus

dsDNA vector

Non-enveloped

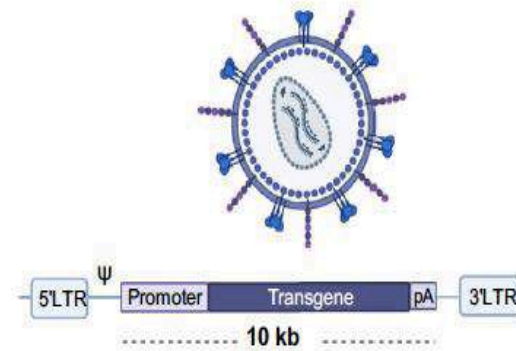
Packaging capacity: up to 36 kb

Mainly episomal

Serotype-dependent pre-existing immunity

Acute inflammatory responses

High liver transduction



Lentivirus

ssRNA vector

Enveloped

Packaging capacity: up to 10 kb

Integrative

Low pre-existing immunity

Poorly characterized in vivo

Ideal for hematopoietic stem cells (HSCs) transduction

Herpes Virus



Herpes virus

dsDNA vector

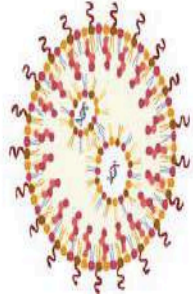
Enveloped

Packaging capacity: 30-50 kb

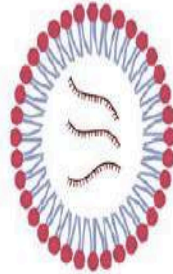
Episomal

Neurotropic virus

Non-viral vectors



Cationic
polymers



Lipid
nanoparticles

Safer with low toxicity

Low immunogenicity

Simpler to use

Allows for large-scale production

Not very efficient in delivering the gene

Expression of the gene is transient

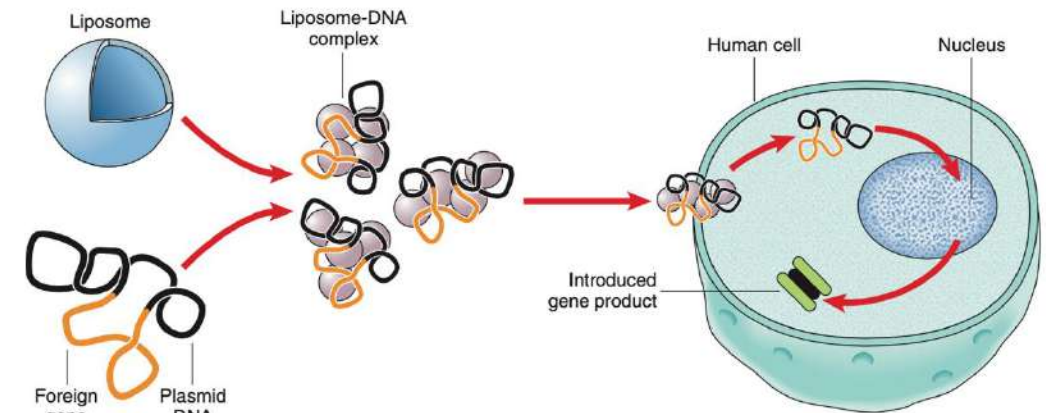


FIGURE 15.9 Diagrammatic representation of liposome-mediated gene therapy.

Image retrieved from: Emery's Elements of Medical Genetics 15th Edition, page 209

Gene Therapy Timeline

*Reading Only

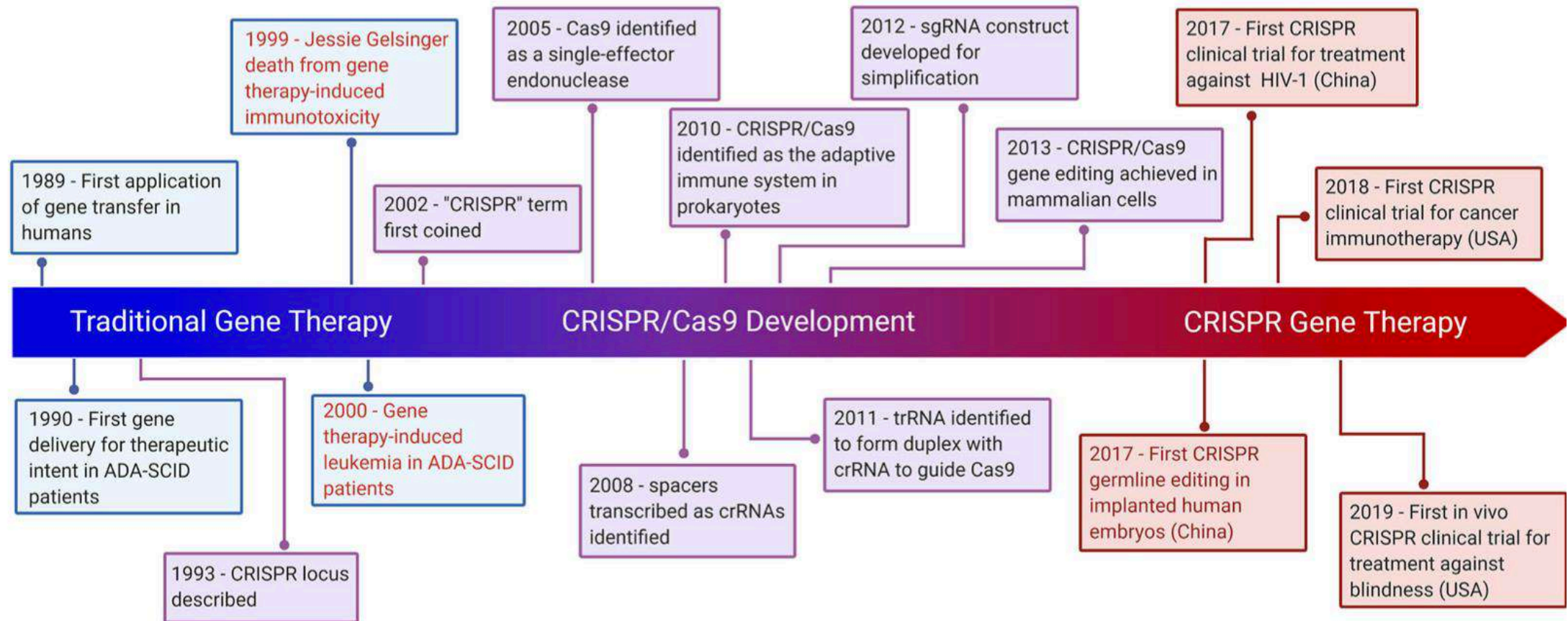
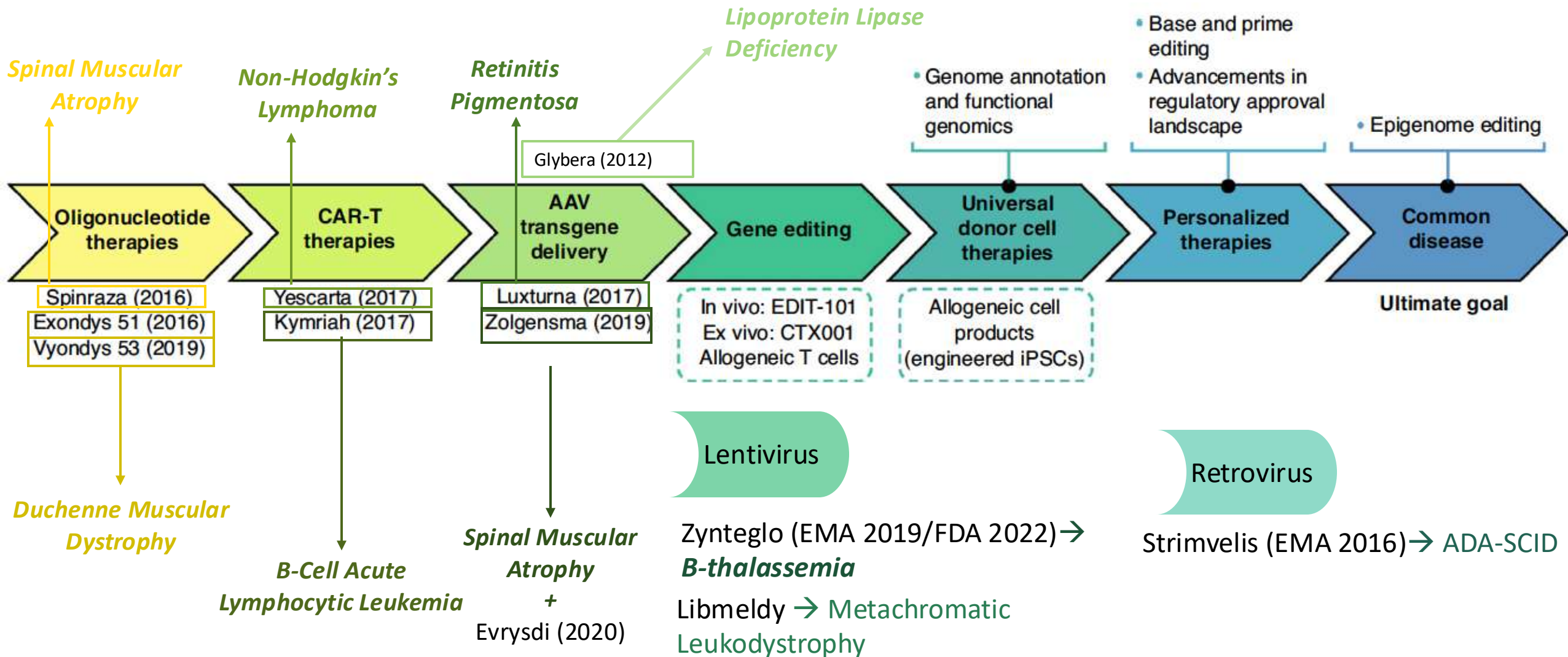
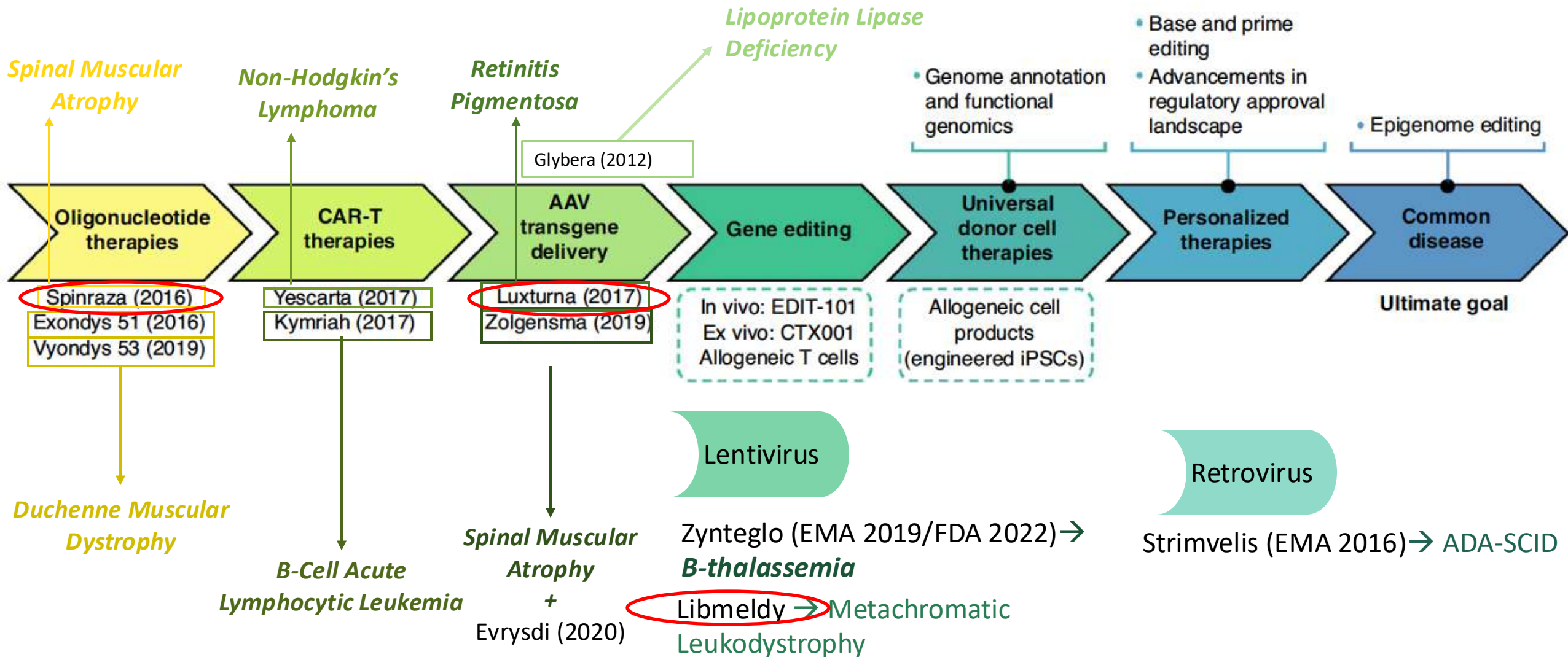


FIGURE 1 | Hallmarks of CRISPR Gene Therapy. Timeline highlighting major events of traditional gene therapy, CRISPR development, and CRISPR gene therapy. text in red denotes gene therapy events which have raised significant ethical concerns.

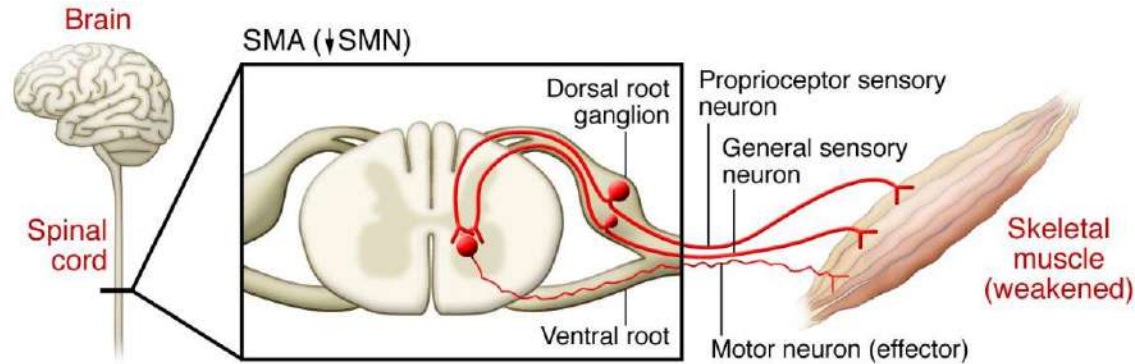
Examples of Gene Therapy Medications:



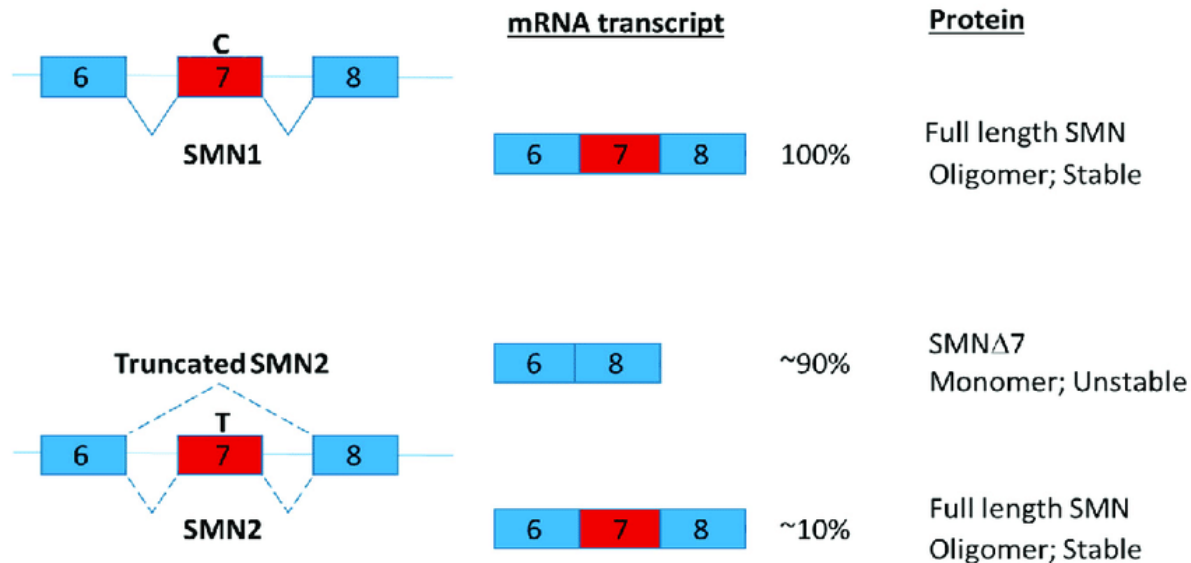
Examples of Gene Therapy Medications:



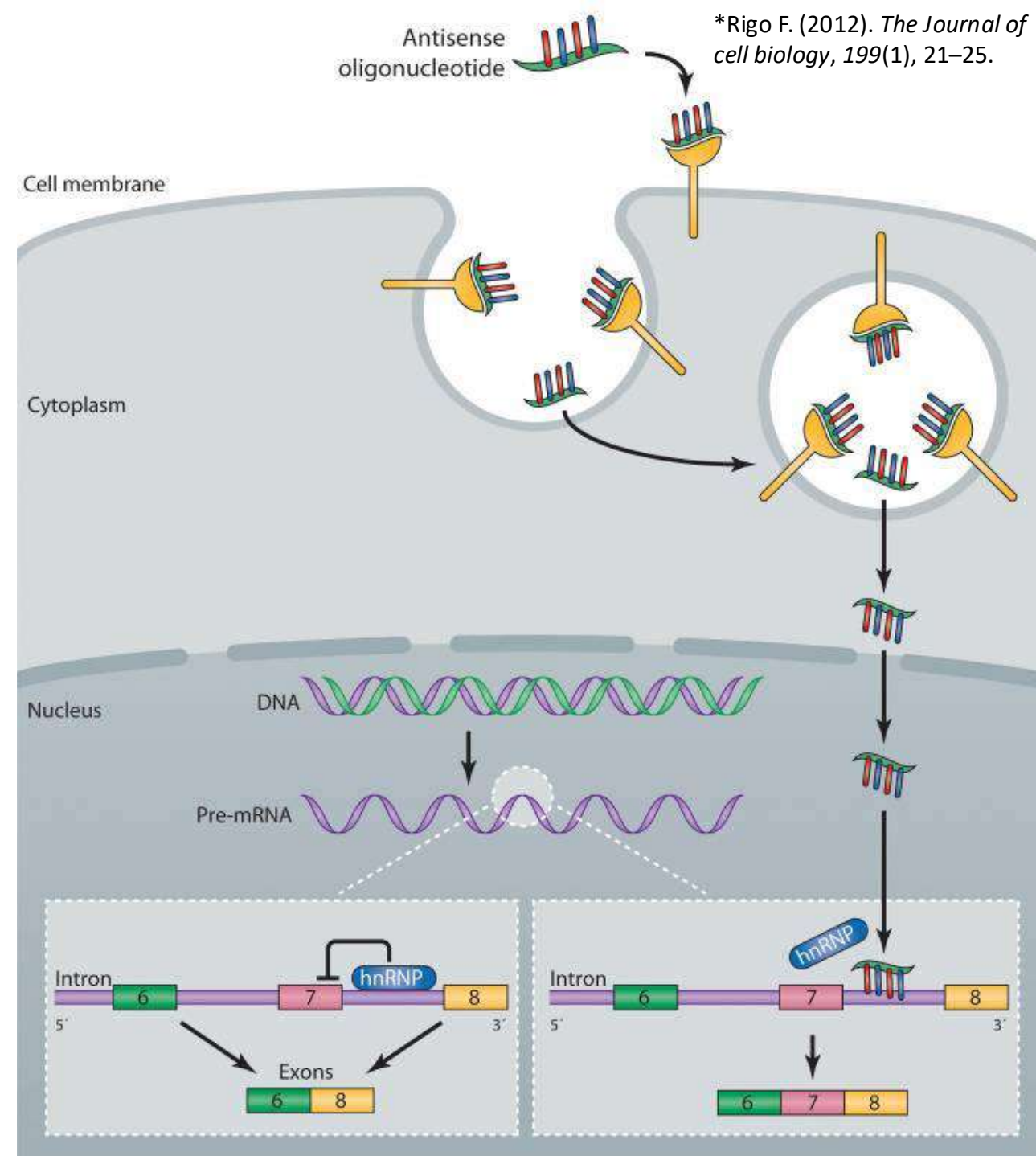
Use of Spinraza in Spinal Muscular Atrophy



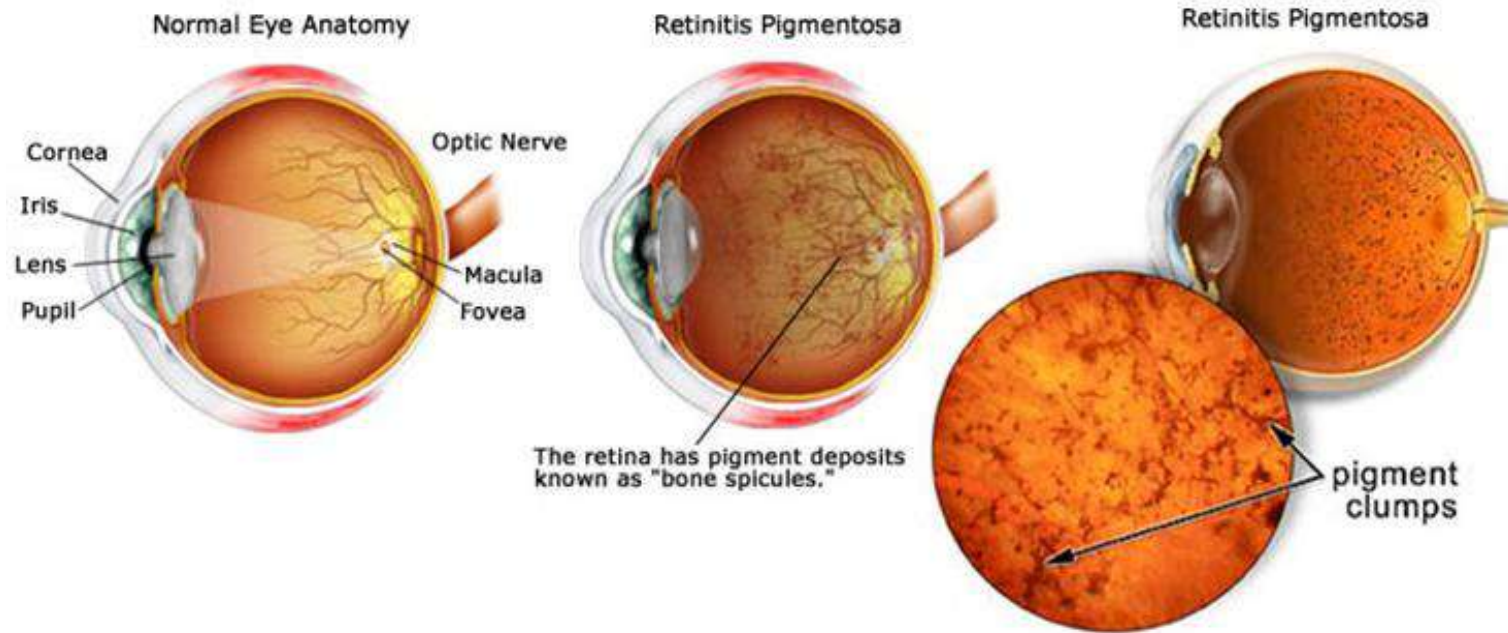
*Swoboda KJ. *The Journal of clinical investigation*, 121(8), 2978–2981.



*Wong CM et al.(2018). *International journal of molecular sciences*, 19(2), 632.



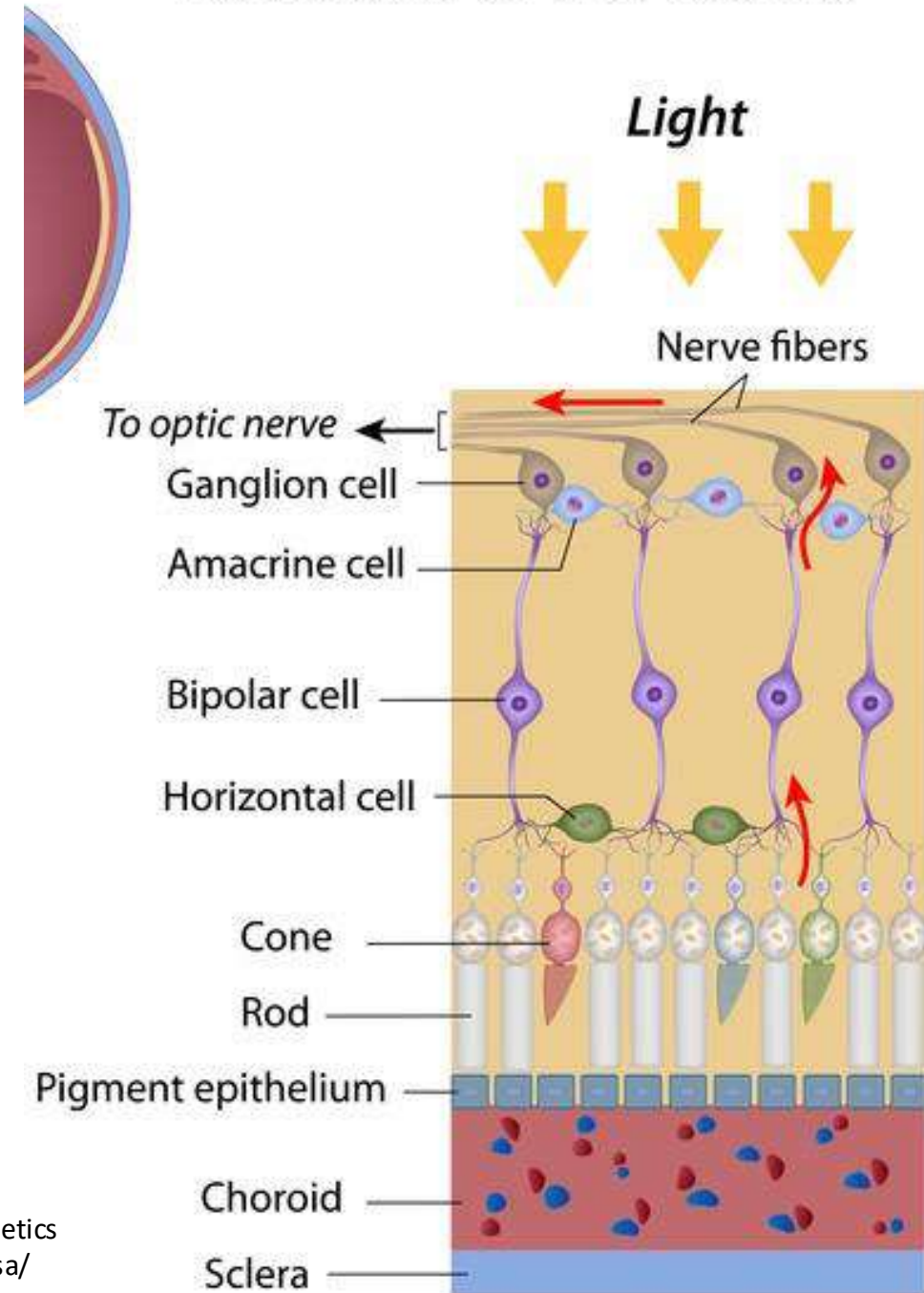
Use of Luxturna in Retinitis Pigmentosa



*<https://irisvision.com/retinitis-pigmentosa-causes-and-symptoms/>

*<https://medlineplus.gov/genetics/condition/retinitis-pigmentosa/>

Structure of the Retina



Use of Luxturna in Retinitis Pigmentosa

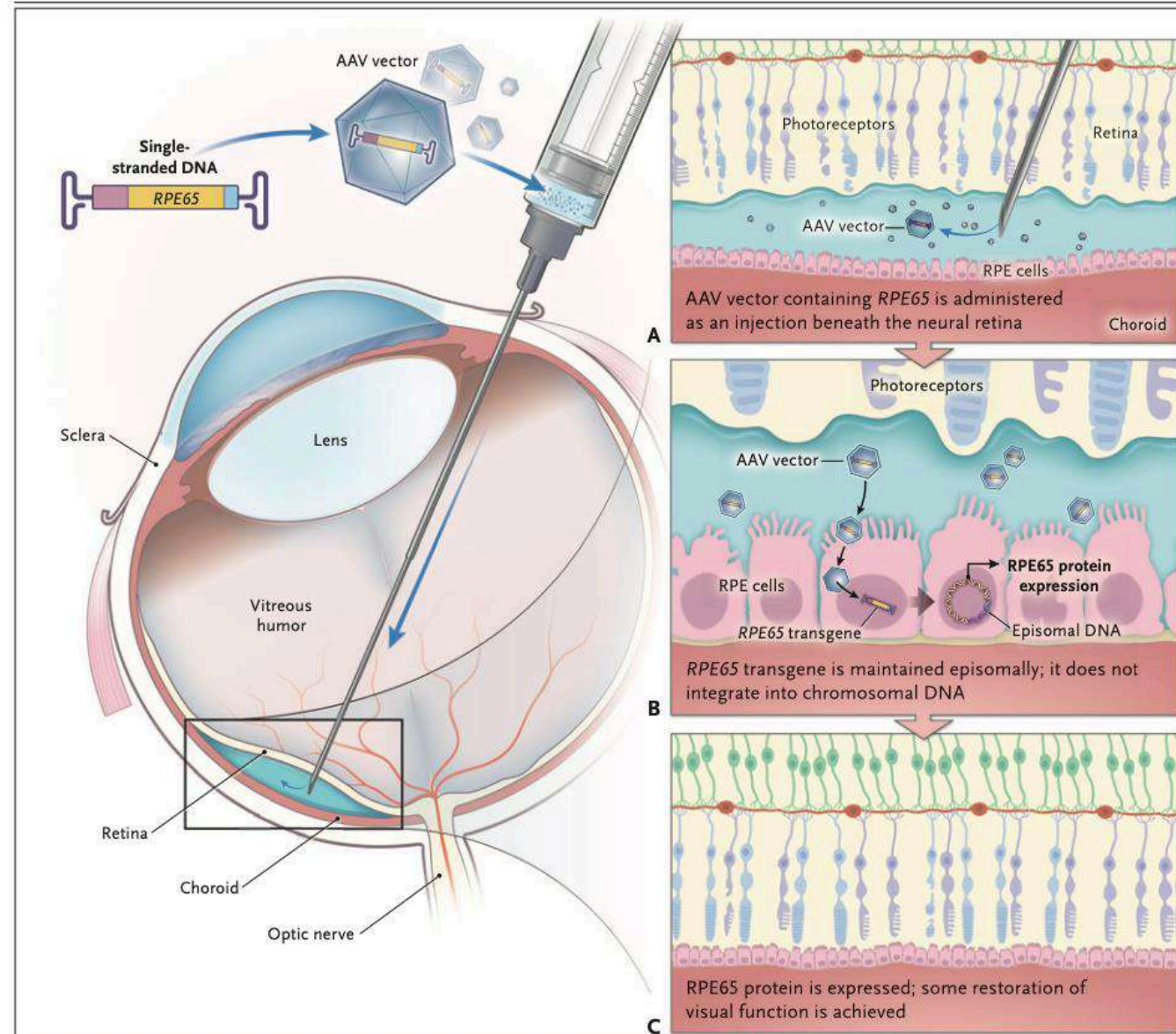
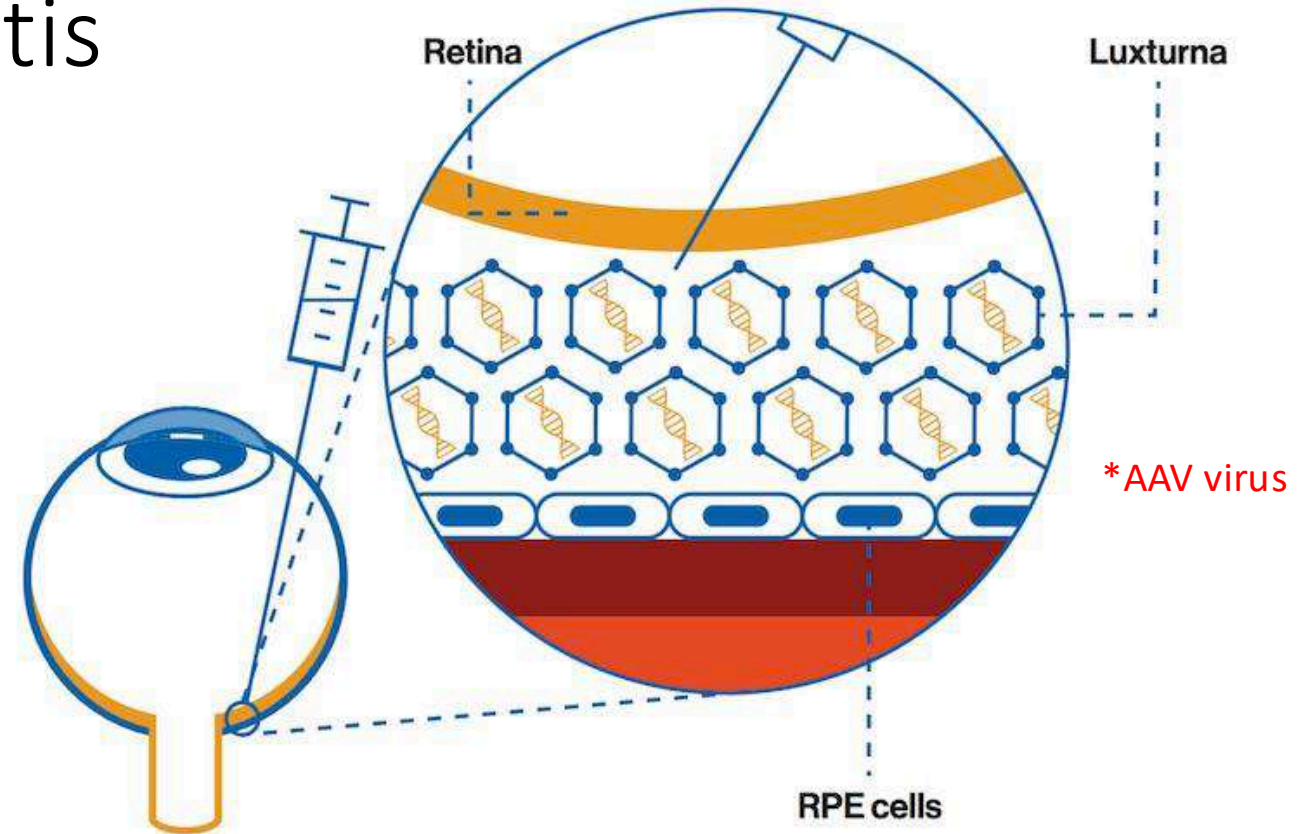
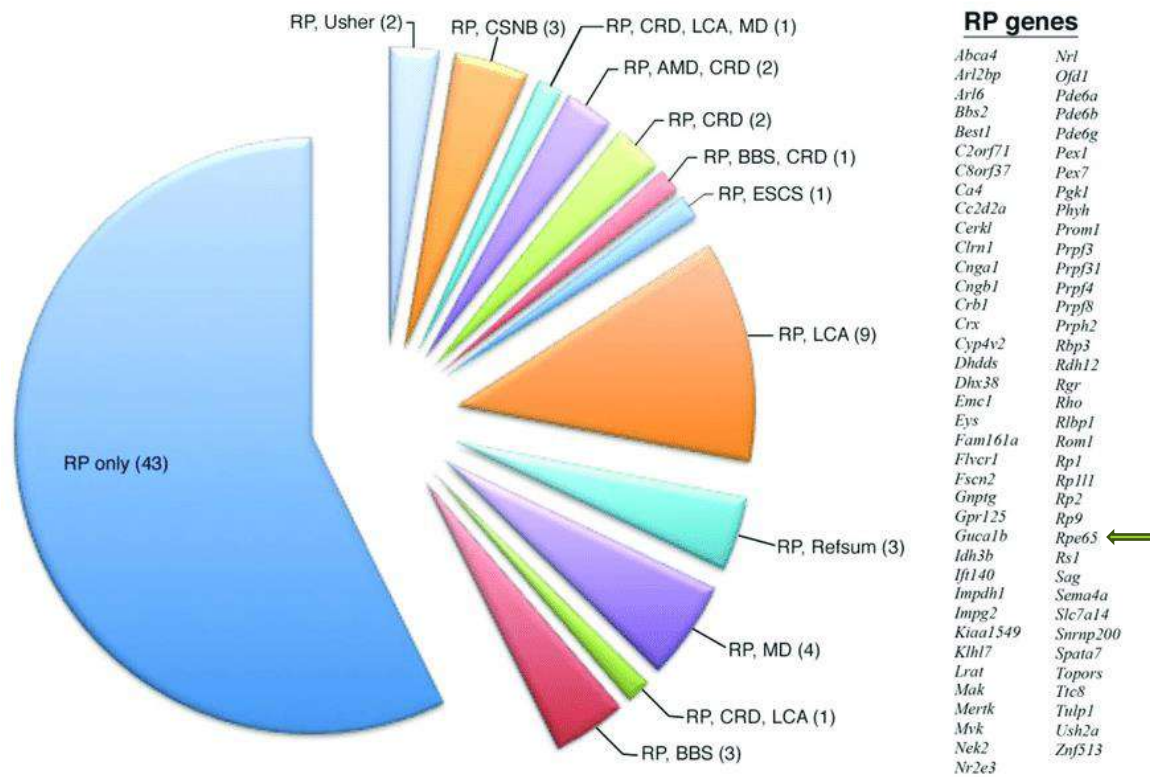


Figure 2. In Vivo Delivery of Gene Therapy.

An example of in vivo gene therapy is the treatment of vision loss caused by loss-of-function variants in *RPE65*, which encodes an enzyme that converts all-trans-retinyl ester to 11-cis-retinol, part of the visual cycle that takes place in the retinal pigment epithelium. The gene is delivered within an adeno-associated viral (AAV) vector by injection beneath the neural retina, through vitrectomy followed by direct injection in an operative procedure. A false space ("bleb") under the retina is created by injecting vector suspended in fluid, whereupon the vector transduces retinal pigment epithelial (RPE) cells. The transgene remains episomal; it does not integrate into the DNA of the cell.

Use of Luxturna in Retinitis Pigmentosa

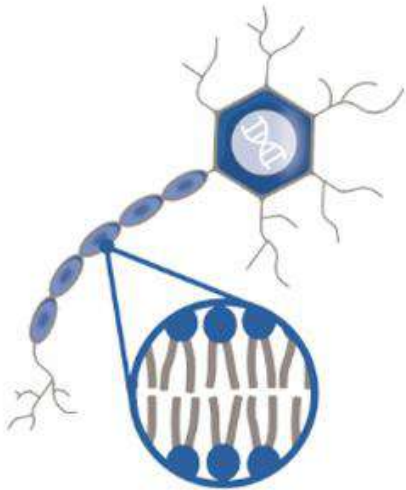


*<https://www.labiotech.eu/trends-news/luxturna-europe-gene-therapy-blindness/>

*Mustafi D et al. (2019). *Pharmaceuticals (Basel, Switzerland)*, 12(2), 74.

Use of Libmeldy in Metachromatic Leukodystrophy

PHYSIOLOGIC



Sulfatides are a major component of the myelin membrane⁴



The **ARSA** enzyme usually breaks down sulfatides in the lysosome^{1,5}

PATHOLOGIC



Mutations in the **ARSA** gene causes deficiency of the ARSA enzyme reducing sulfatide breakdown



Intra-lysosomal **sulfatides accumulation** leads to demyelination and neurodegeneration^{1,4-6}

Use of Libmeldy in Metachromatic Leukodystrophy

STEP 3

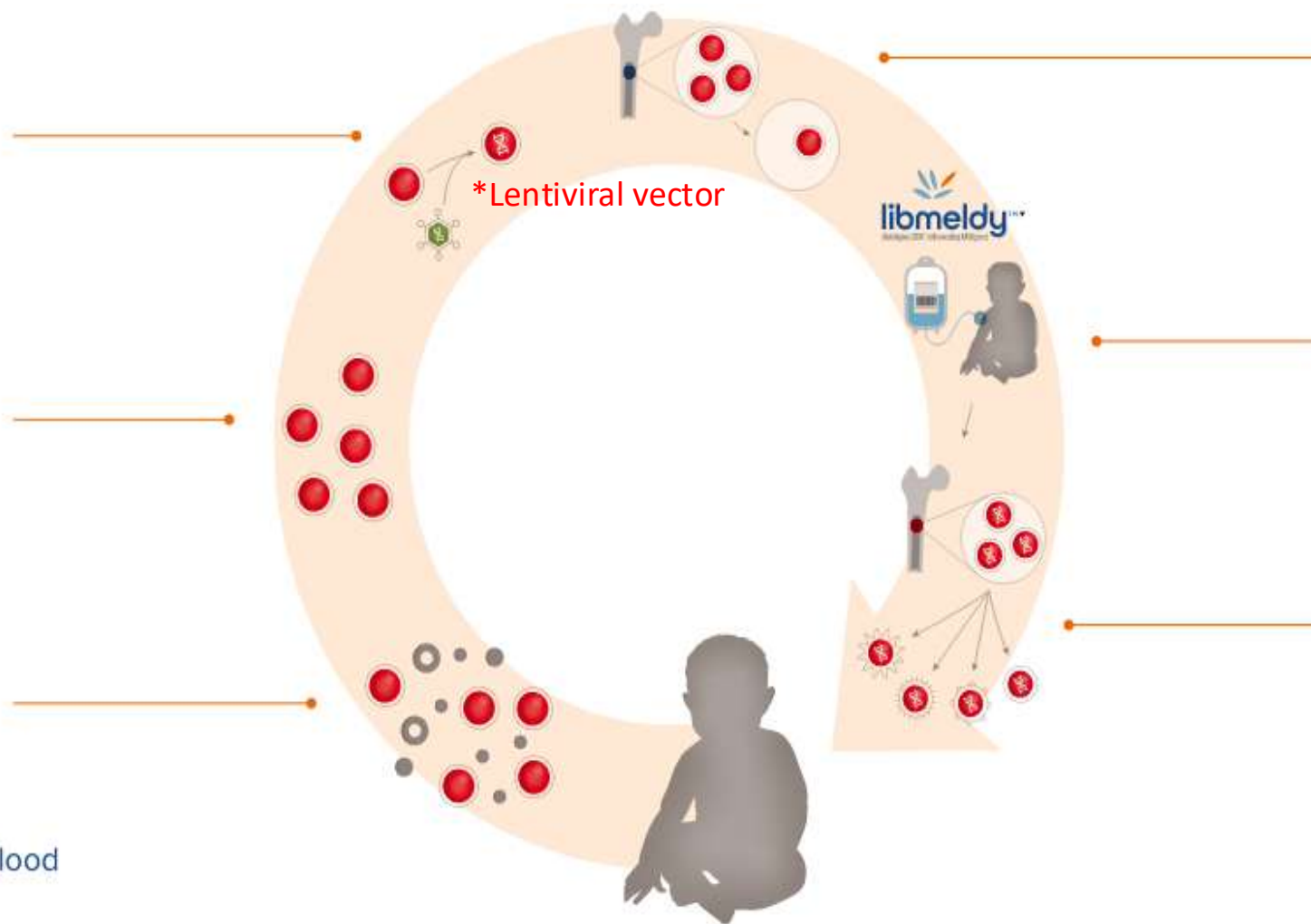
A working copy of the gene is inserted into the cells

STEP 2

Blood stem cells are selected and purified

STEP 1

Cells are collected from the patient's bone marrow or mobilized peripheral blood



STEP 4

Patient is conditioned to receive Libmeldy

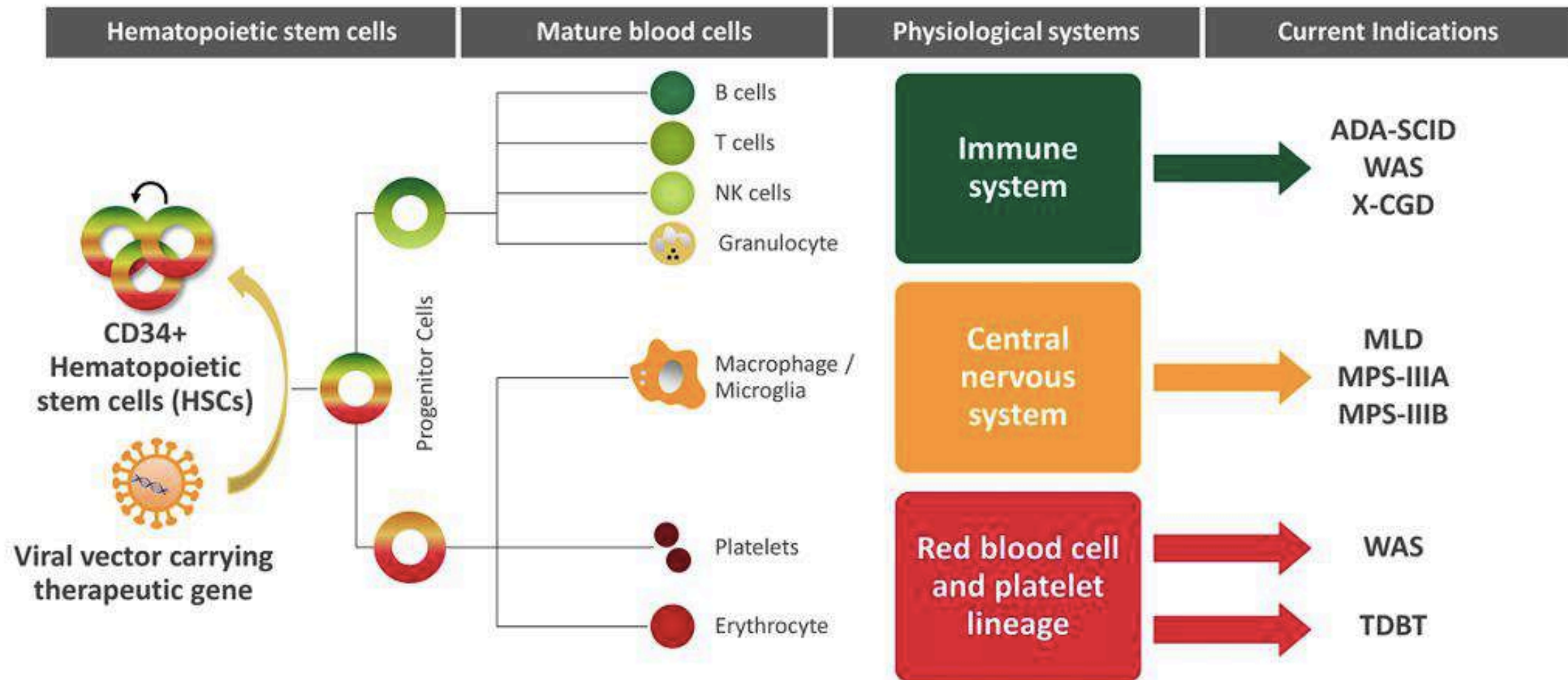
STEP 5

Gene-corrected cells are infused into the patient

STEP 6

Cells engraft and start to proliferate

Delivering Therapeutic Genes to Multiple Physiological Systems



Potential for sustained disease correction after a single administration via gene-modified HSCs engraftment

Gene Therapy for Treatment of Cancer

The delivery of genetic elements to the cancer cell or to the cells of the immune response

- to correct the abnormalities in the cancer tissue
- to induce an immune response against the cancer cells

The corrective strategies can involve:

- Replacing missing or defective tumor suppressor genes
- Suppressing the action of oncogenes
- Programming normal or cancer cells to release into the systemic circulation molecules which suppress the growth of cancer cells or their vasculature
- Inserting genes to cancer cells that make them more sensitive to chemotherapy and radiotherapy
- Suicide gene therapy: Inserting suicide genes into cancer cells with drugs only activated with these genes

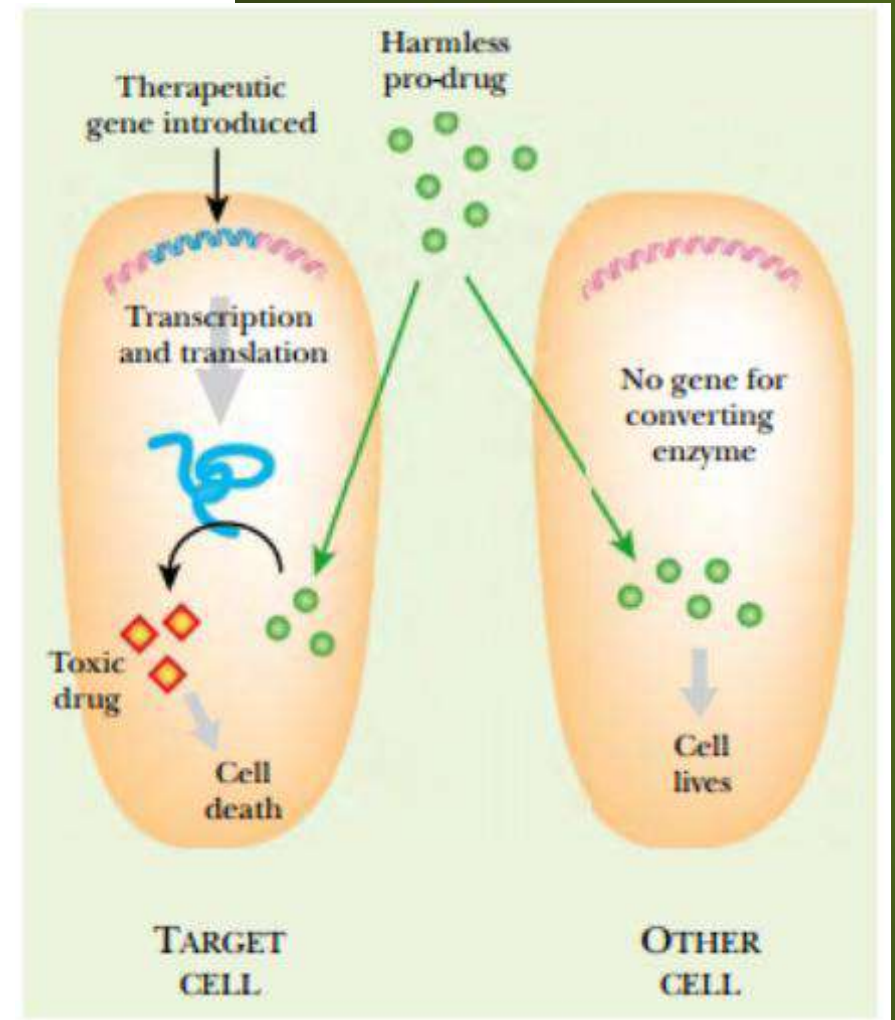
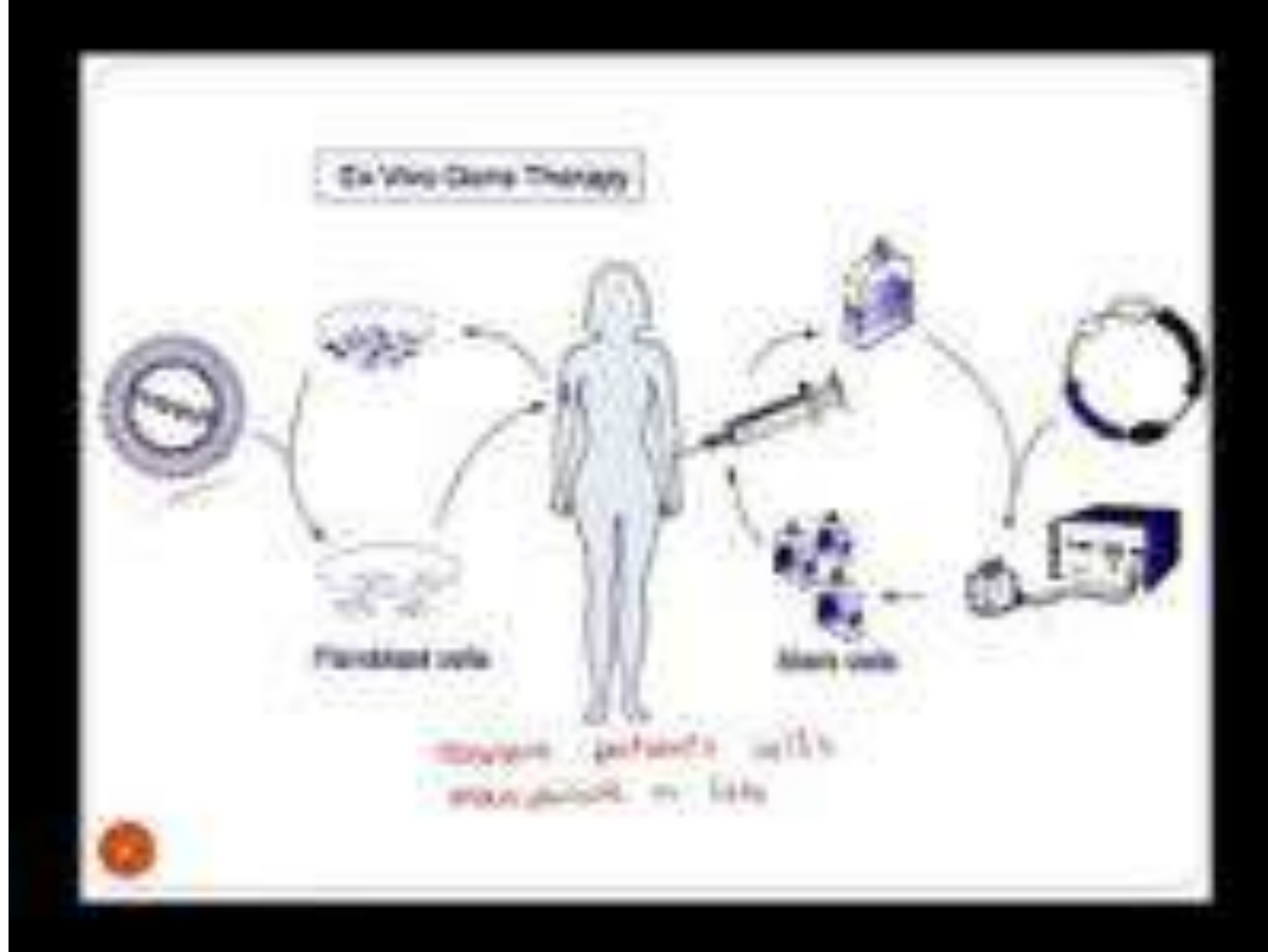


Image retrieved from <https://istudy.pk/aggressive-gene-therapy-cancer/>

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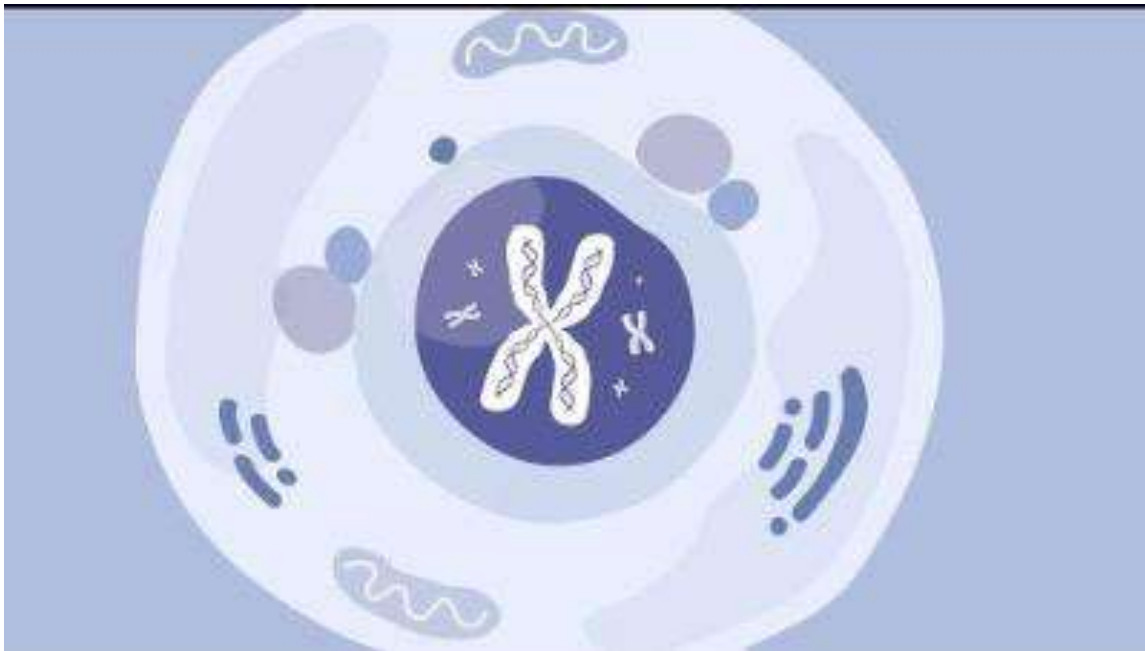


<https://www.youtube.com/watch?v=Z1MR0o4O0-4>

What were the two mentioned reactions that caused setbacks to the field of Gene Therapy?

Gene Therapy Basics

- <https://youtu.be/https://www.youtube.com/watch?v=kAtd9X29SdQ>



AMERICAN SOCIETY of
**GENE & CELL
THERAPY**

CRISPR-Cas9

- <https://www.youtube.com/watch?v=QA0FdR-wmYI>

 AMBOSS

MOLECULAR BIOLOGY

CRISPR-Cas

